

Technologies for E-Commerce

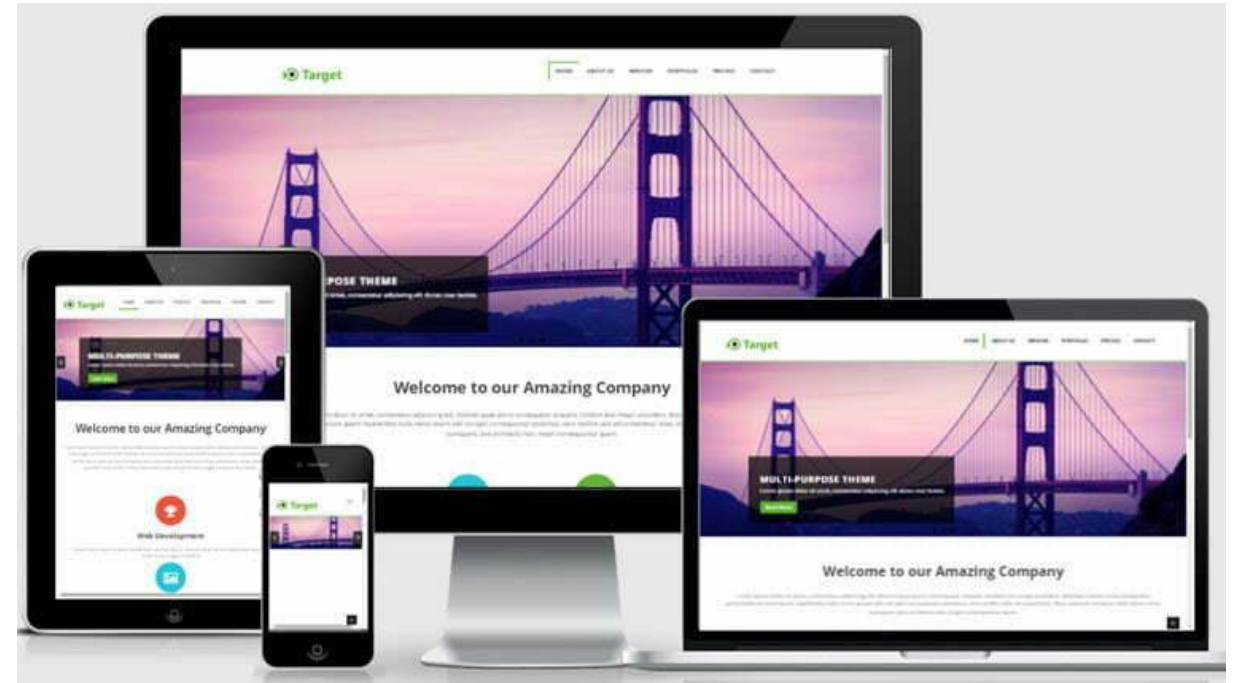
CAN302

**Department of Communications and Networking
Xi'an Jiaotong-Liverpool University (XJTLU)**

Lecture 3 – Insight into JS & Bootstrap

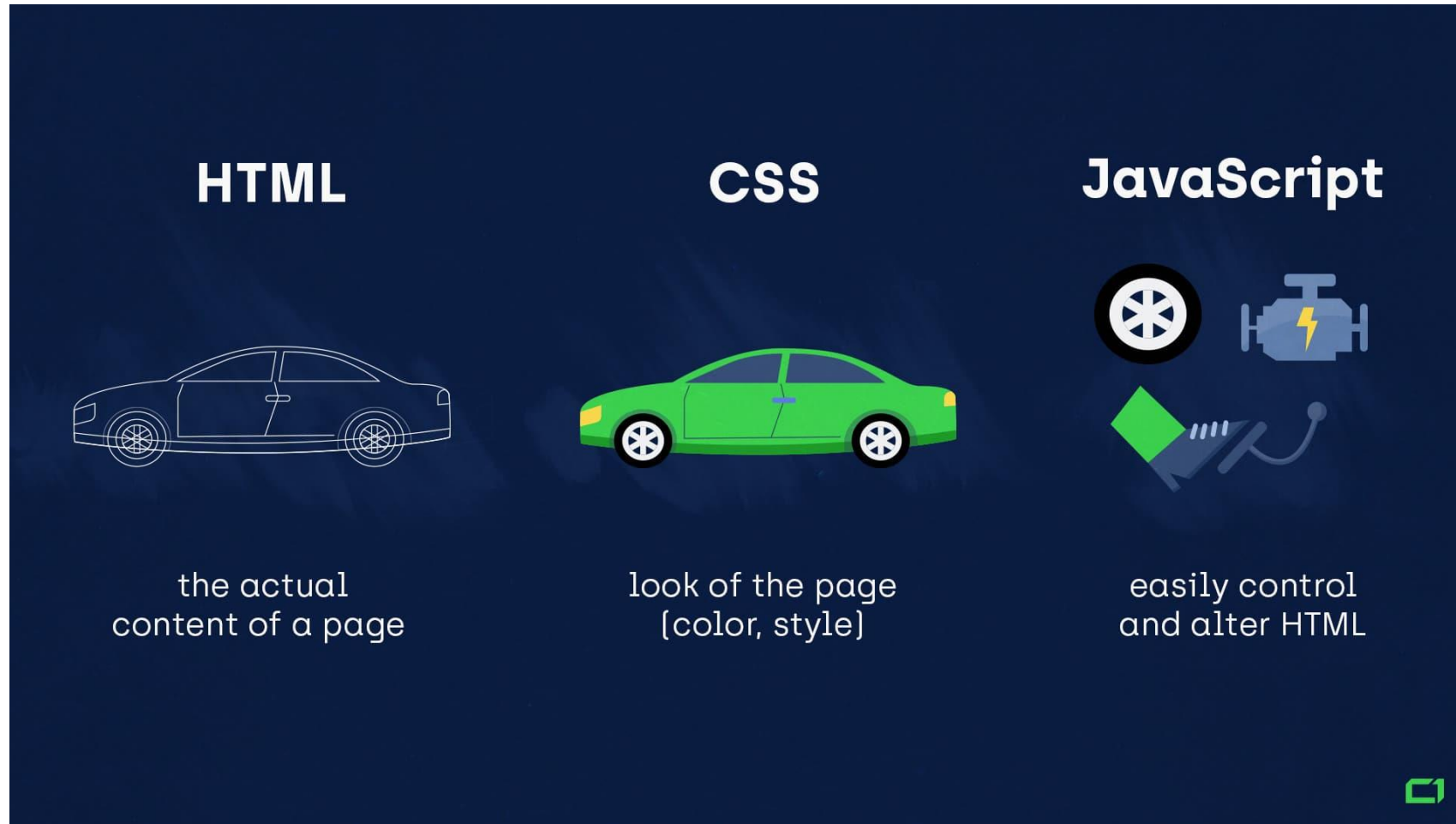


Outline: @JavaScript



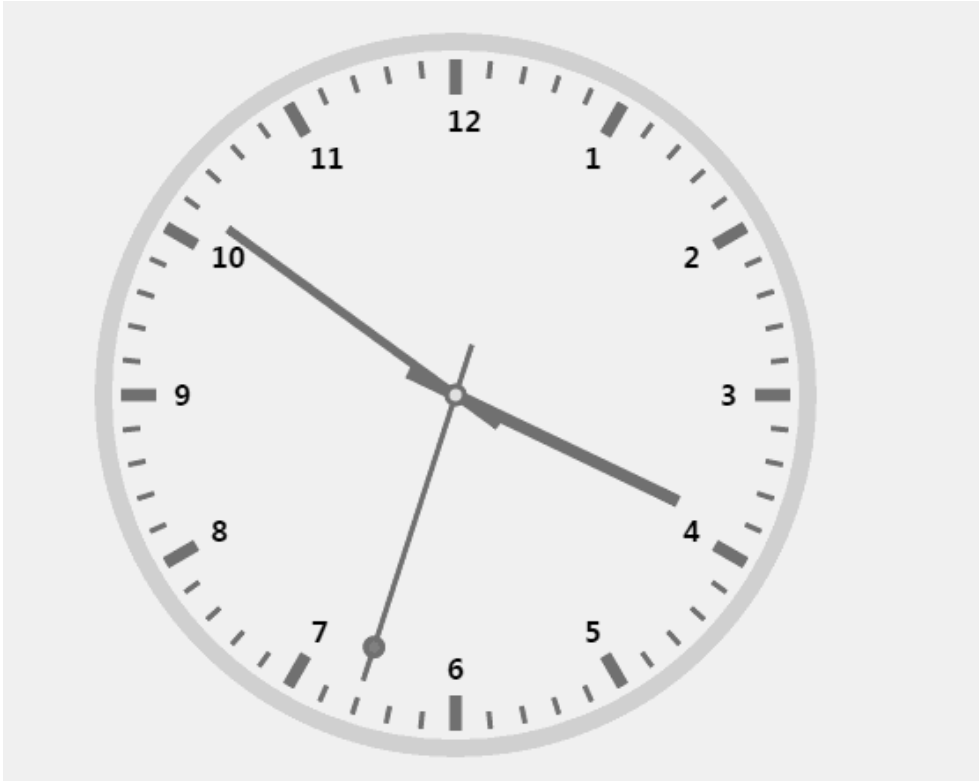
@Bootstrap

JavaScript (JS)



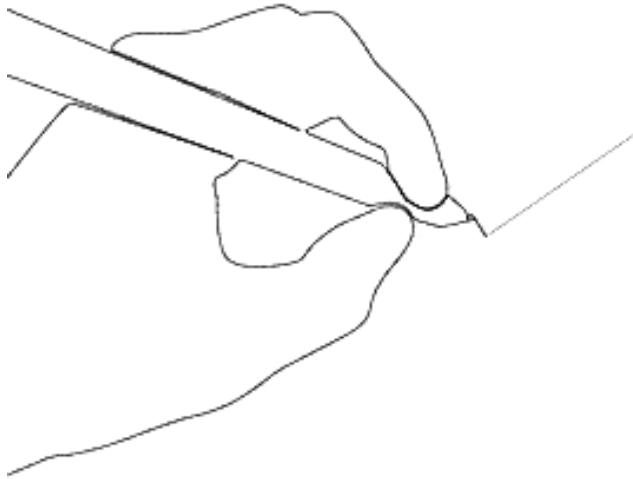
- JavaScript is a powerful programming language that can add **interactivity** to a website.

A JS sample – H5 clock



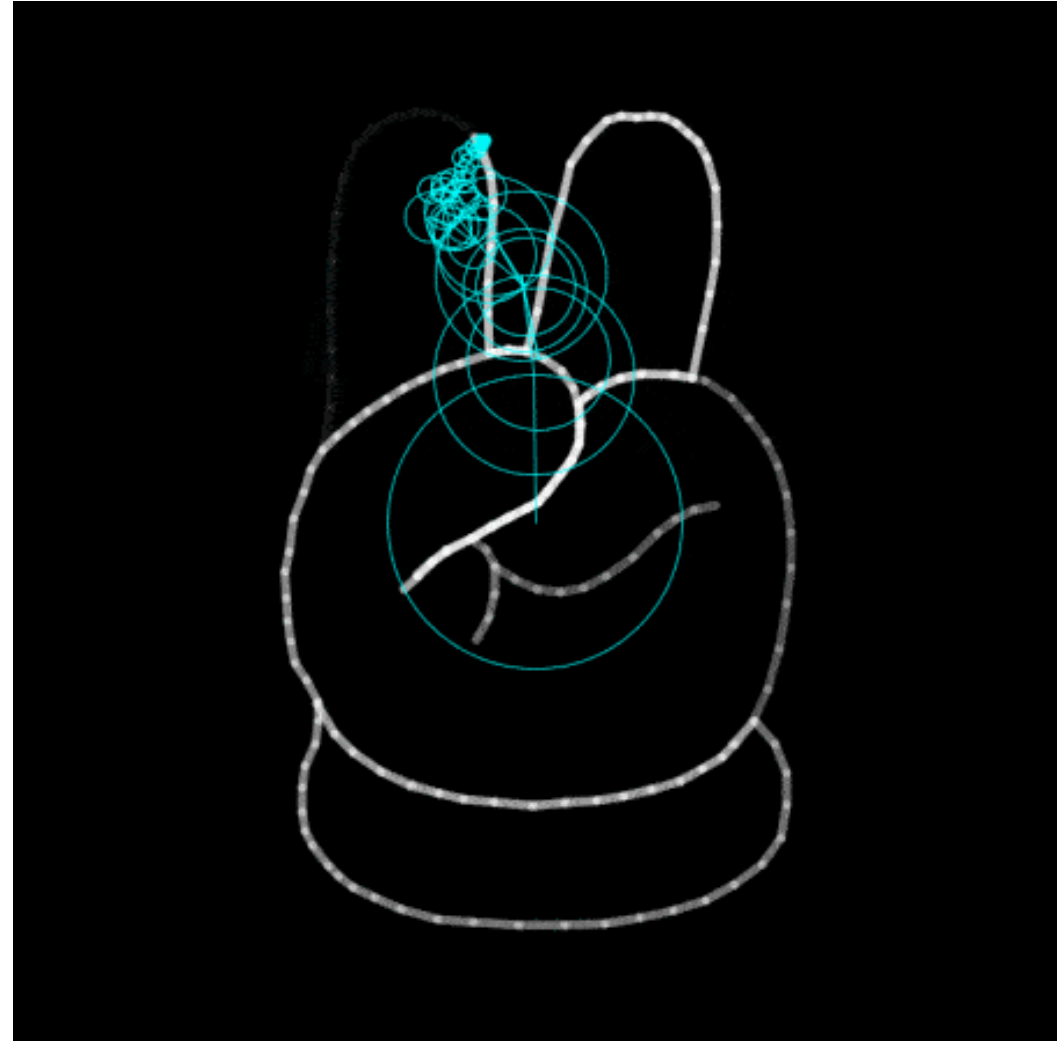
- HTML5 has the `<canvas>` tag, it allowed to draw anything on it.
- A demo clock dynamically drawn by JavaScript.

Draw something cool



$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{-j2\pi\mu t} d\mu$$

- Ah ~ Fourier Transform



<https://openprocessing.org/sketch/758144>

Even mining Bitcoin!

The image is a screenshot of the COINIMP website. At the top, the COINIMP logo is on the left, and links for 'Documentation', 'Referral Program', 'Log in', and 'Sign Up' are on the right. The main heading reads 'FREE JavaScript Mining' in blue, followed by 'Use our Monero JavaScript web miner and EARN MONEY with your page traffic!' in black. Below this, a paragraph explains that users will enjoy an ad-free experience and that the script is completely free, with a 1% fee being refunded through the referral program. A blue button labeled 'START MINING NOW!' is centered. Further down, the section 'What is CoinImp?' is followed by a description of the miner as an alternative to ads. A blue cartoon bat character is visible on the right side of the page.

COINIMP Documentation Referral Program Log in Sign Up

FREE JavaScript Mining

Use our Monero JavaScript web miner and **EARN MONEY** with your page traffic!

Your users will enjoy an ad-free experience when running the script in their browsers while they mine cryptocurrency for you. Unique offer on the market - completely free script for web miners! We do take 1% fee, but we give you this back (and more!) in form of [Referral Program](#)! So this is powerful solution for JavaScript mining at effective **0%** of your total hash rate!

START MINING NOW!

What is CoinImp?

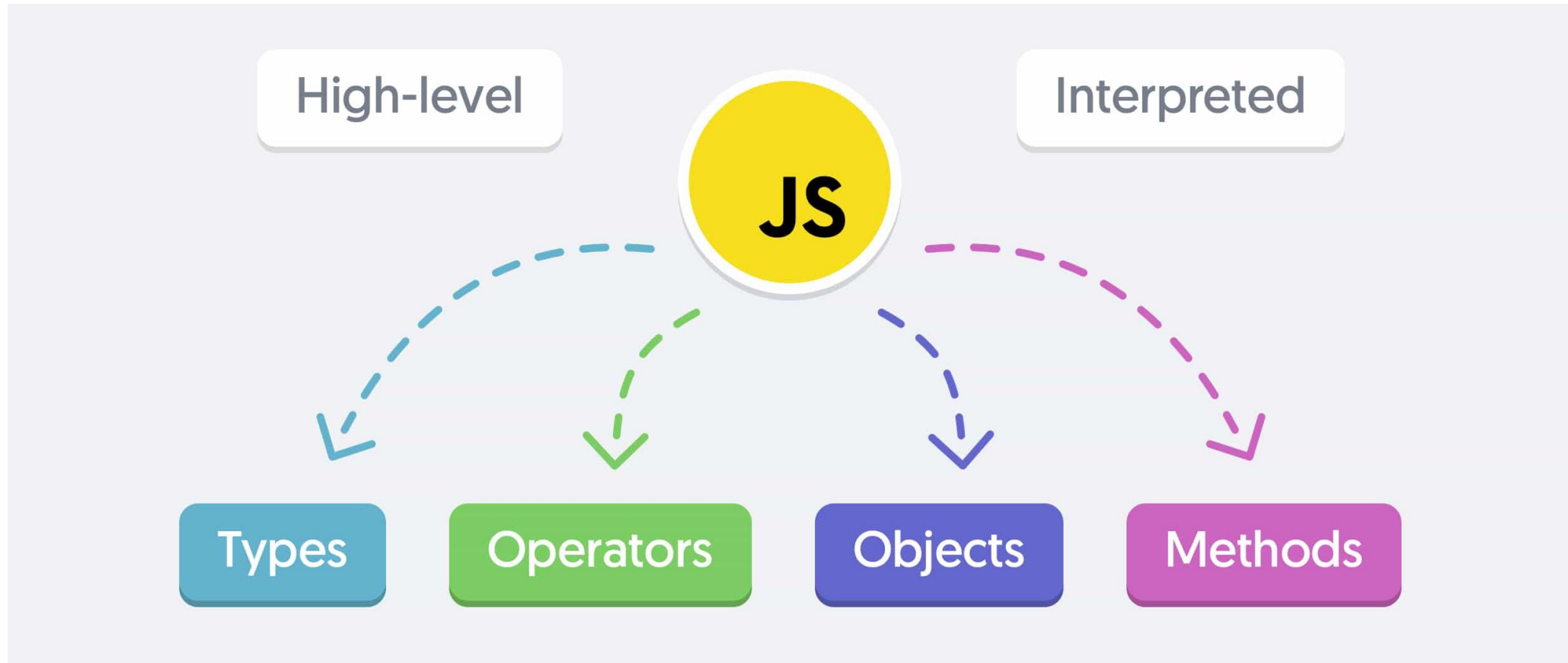
CoinImp is the new effective JavaScript miner that can be easily embedded in your website. It's an excellent alternative to annoying ads. This way users will pay you with their CPU power for accessing your content.



- It is **NOT** legal and **NEVER** do it!

<https://www.geckoandfly.com/26702/cryptocurrency-mining-javascripts-mine-bitcoin-online/>

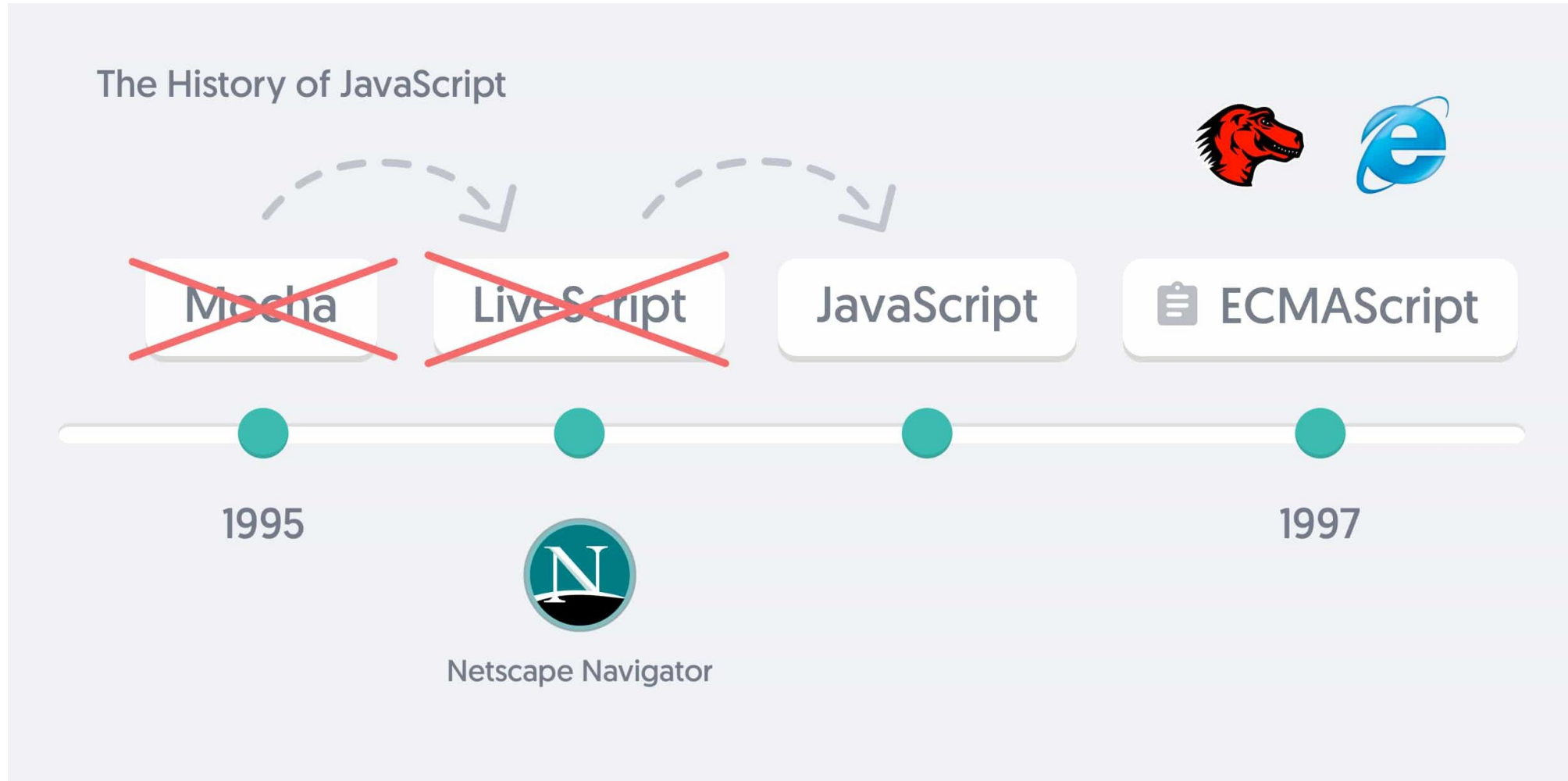
What is JS?



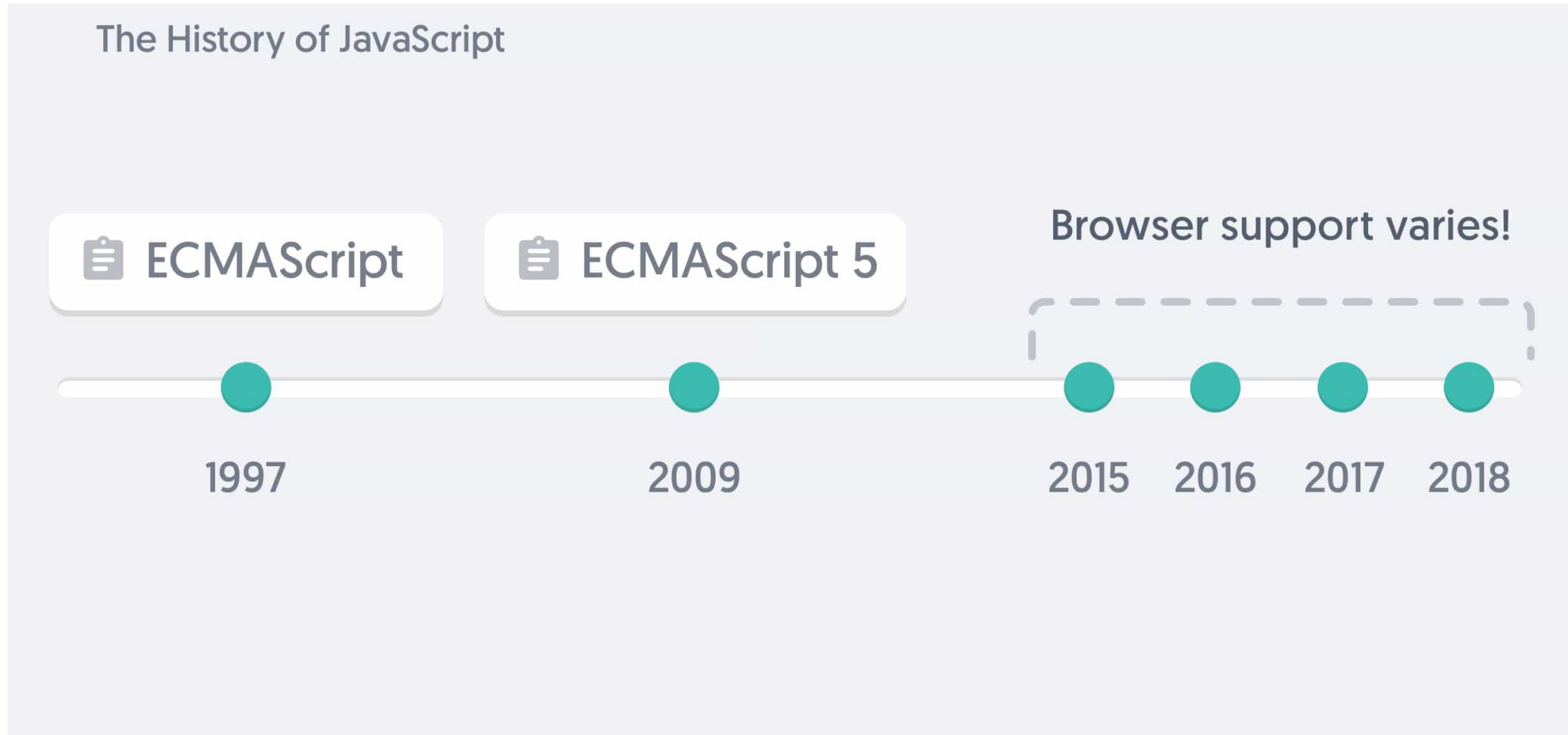
- JavaScript is a high level and **interpretive** programming language. **Is it true?**
- JavaScript has types, operators, standard built-in objects and methods.

<https://ultimatecourses.com/blog/an-introduction-to-javascript>

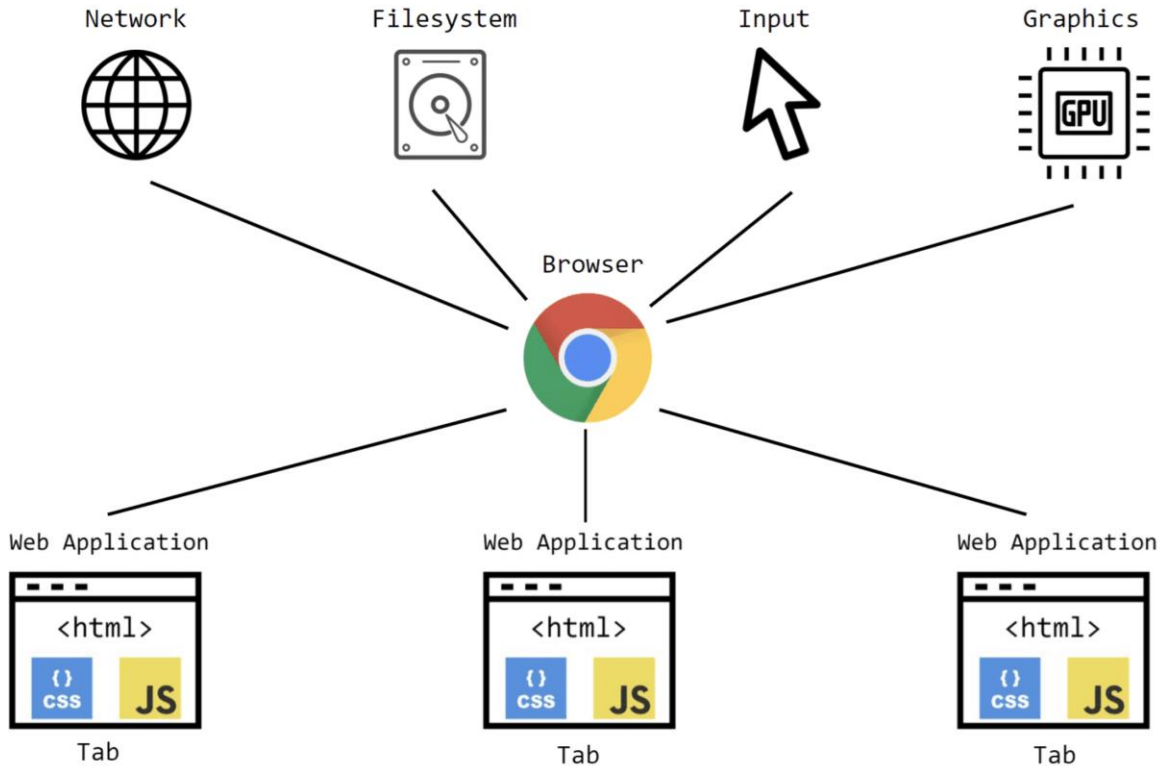
Mocha → LiveScript → JavaScript: JS



Mocha → LiveScript → JavaScript: JS



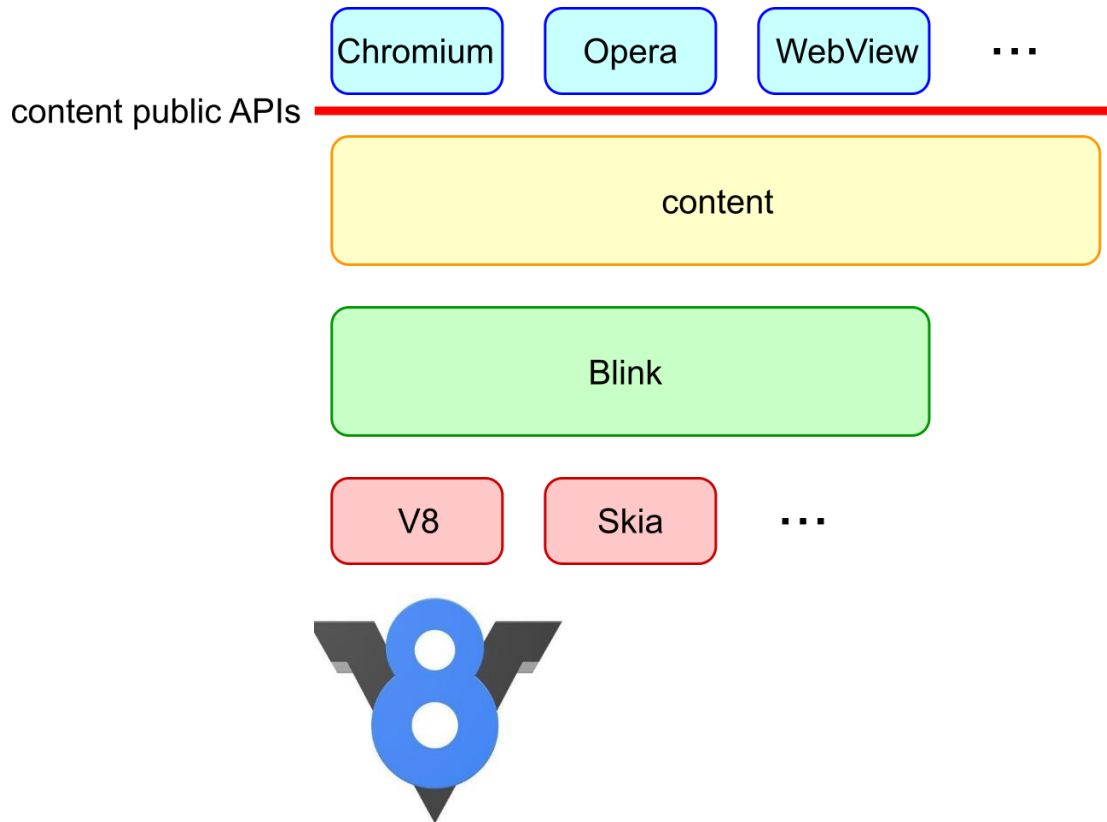
The modern browser



<https://webperf.tips/tip/browser-process-model/>

- A *Web Browser* is a compiled binary (i.e. chrome.exe) that is installed on and launched from a user's device.
- It's responsible for interfacing with a variety of sources. Some examples include:
- The device's filesystem, for loading user preferences, history, favorites, etc.
- The network, to load and deliver remote data
- The OS's input events, such as clicking and typing
- The GPU and display, to present pixels on-screen
- Remote code loaded for a web application, such as HTML, JavaScript, and CSS files defined by a web application

The structure of Browser



- Blink is a rendering engine of the web platform. Roughly speaking, Blink implements everything that renders content inside a browser tab:
- Implement the specs of the web platform (e.g., HTML standard), including DOM, CSS and Web IDL
- **Embed V8 and run JavaScript**
- Request resources from the underlying network stack
- Build DOM trees
- Calculate style and layout
- Embed Chrome Compositor and draw graphics
- Blink is embedded by many customers such as Chromium, Android WebView and Opera via content public APIs.

V8 engine history



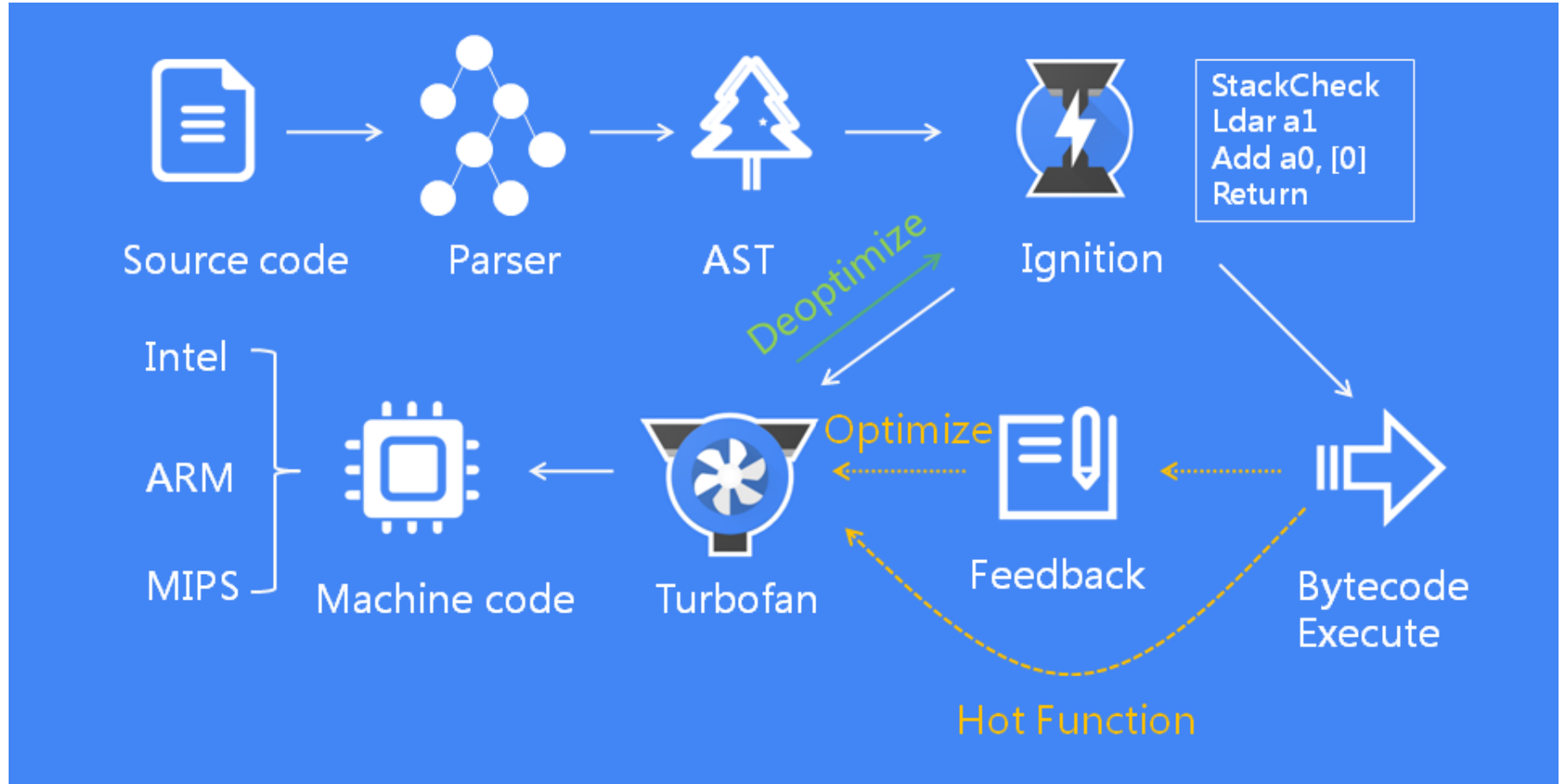
- V8 was the first really fast **JavaScript Virtual Machine**
 - Launched with Chrome in 2008
 - 10x faster than competition at release
 - 10x faster today than in 2008
- 2008 - Full-Codegen
 - Fast AST-walking **JIT compiler** with inline caching
- 2010 - Crankshaft
 - Optimizing JIT compiler with type feedback and deoptimization
- 2015 - TurboFan
 - Optimizing JIT compiler with type and range analysis, sea of nodes

https://docs.google.com/presentation/d/1HgDDXBYqCJNasBKBDf9szap1j4q4wnSHhOYpaNy5mHU/edit#slide=id.g17d335048f_1_1903

Bytecode and Machine code

NO.	Byte Code	Machine Code
01.	Byte Code consisting of binary, hexadecimal, macro instructions like (new, add, swap, etc) and it is not directly understandable by the CPU.	Machine code consisting of binary instructions that are directly understandable by the CPU.
02.	Byte code is considered as the intermediate-level code.	Machine Code is considered as the low-level code.
03.	Byte code is a non-runnable code generated after compilation of source code and it relies on an interpreter (e.g. Virtual Machine) to get executed.	Machine code is a set of instructions in machine language or in binary format and it is directly executed by CPU.
04.	Byte code is executed by the virtual machine then the Central Processing Unit.	Machine code is not executed by a virtual machine it is directly executed by CPU.
05.	Byte code is less specific towards machine than the machine code.	Machine code is more specific towards machine than the byte code.
06.	It is platform-independent as it is dependent on the virtual machine and the system having a virtual machine can be executed irrespective of the platform.	It is not platform independent because the object code of one platform can not be run on the different Operating System. Object varies depending upon system architecture and native instructions associated with the machine.

V8 workflow (with Turbofan)



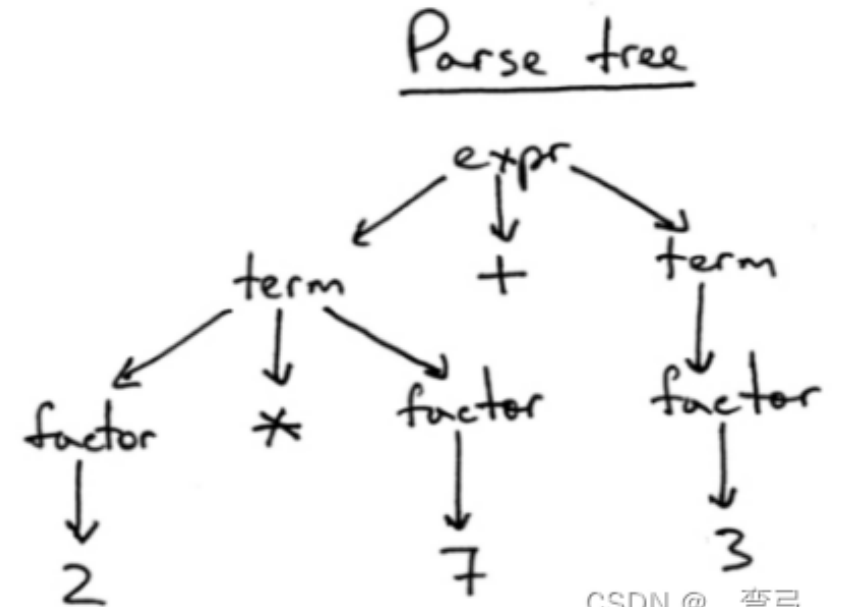
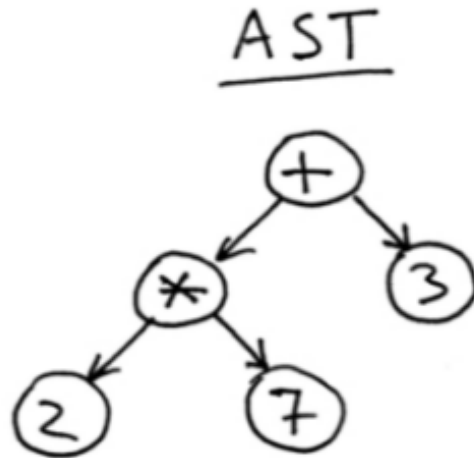
<https://www.bilibili.com/video/BV1zV411z7RX/>

<https://blog.csdn.net/xgangzai/article/details/140860449>

Abstract Syntax Tree and Parse Tree



$2 * 7 + 3$



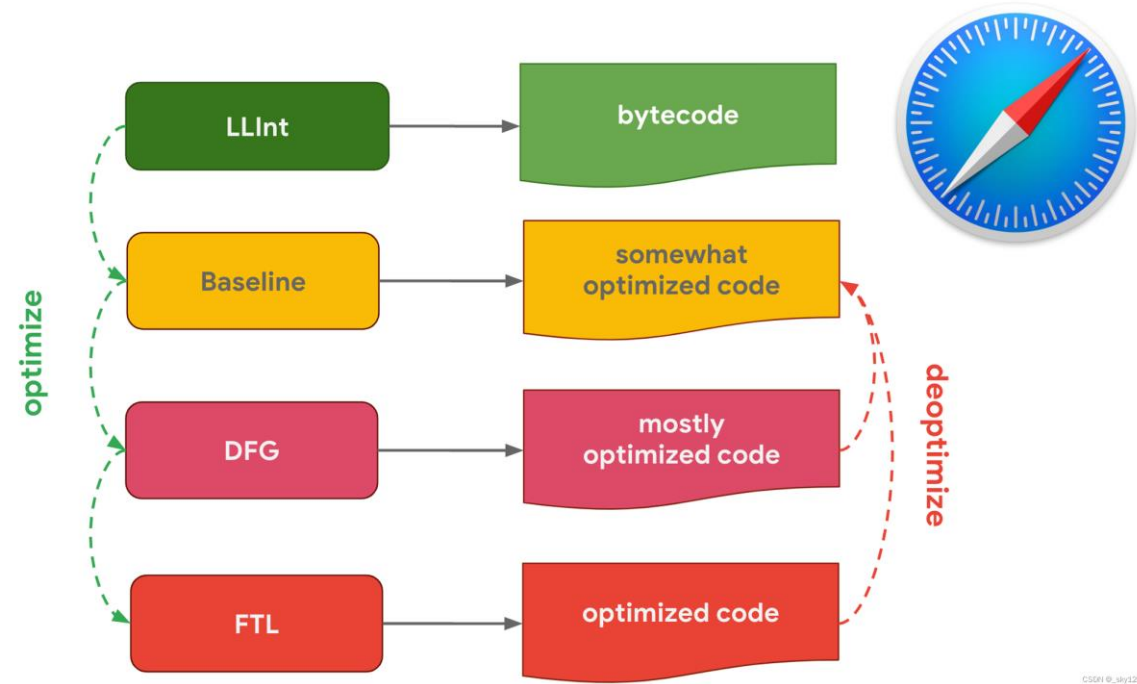
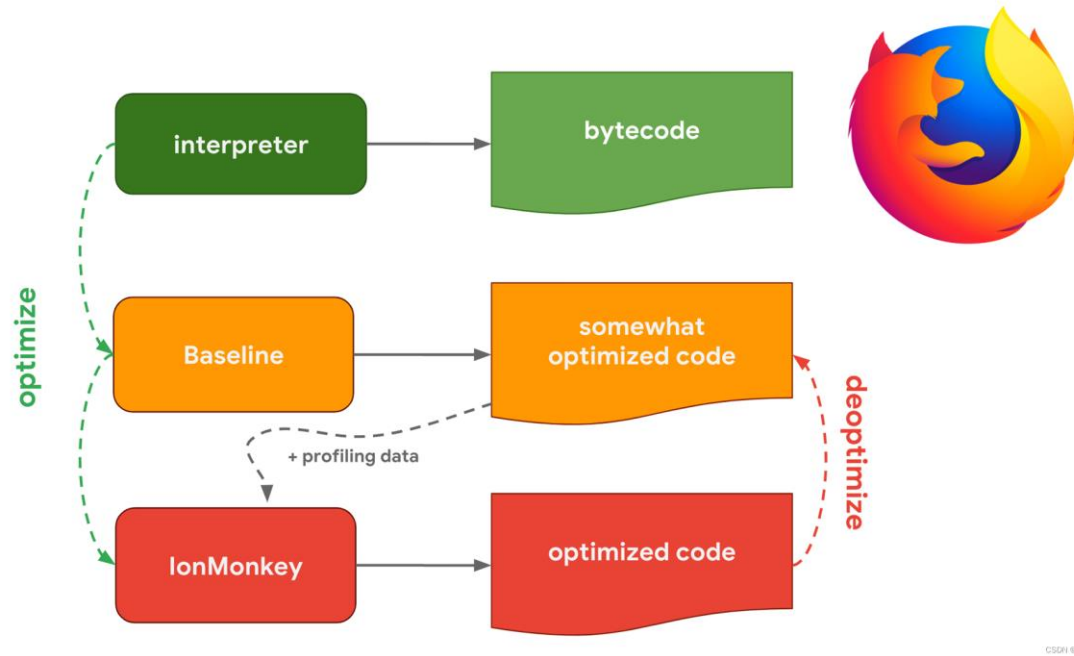
CSDN @__弯弓__

Just in Time - JIT



- JIT (Just-In-Time Compilation) is a compilation process in which code is translated from an intermediate representation or a higher-level language (e.g., JavaScript or Java bytecode) into machine code at runtime, rather than prior to execution. This approach combines the benefits of both interpretation and ahead-of-time (AOT) compilation.
- JIT compilers typically continuously analyze the code as it is executed, identifying parts of the code that are executed frequently (hot spots). If the speedup gains outweigh the compilation overhead, then the JIT compilers will compile those parts into machine code. The compiled code is then executed directly by the processor, which can result in significant performance improvements.

JS engine of Firefox and Safari

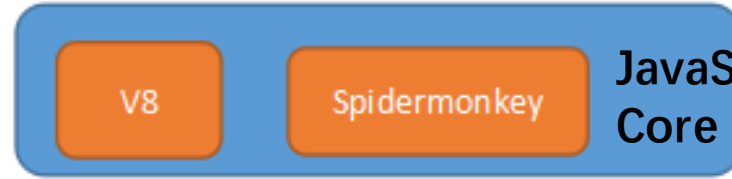


https://blog.csdn.net/qq_45323960/article/details/130124693

Overall of browsers



JavaScript
Core



JavaScript
Core

Javascript
Engine



Render Engine



Browser



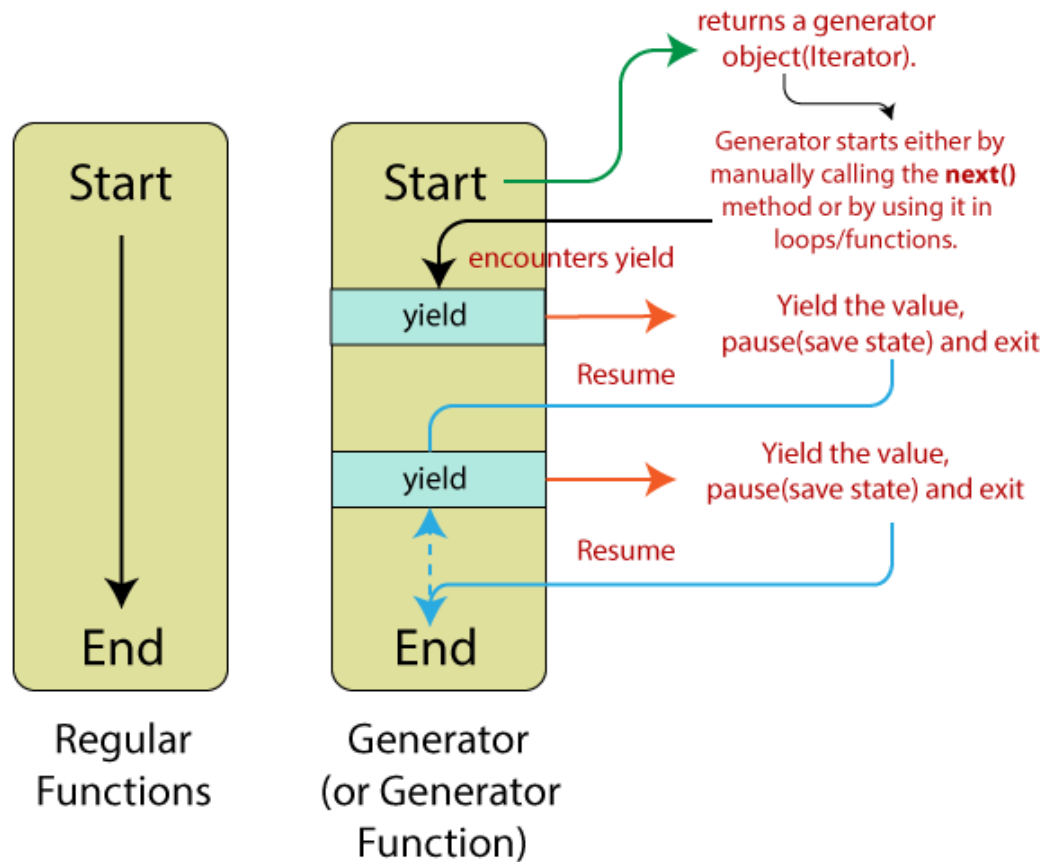
Browser



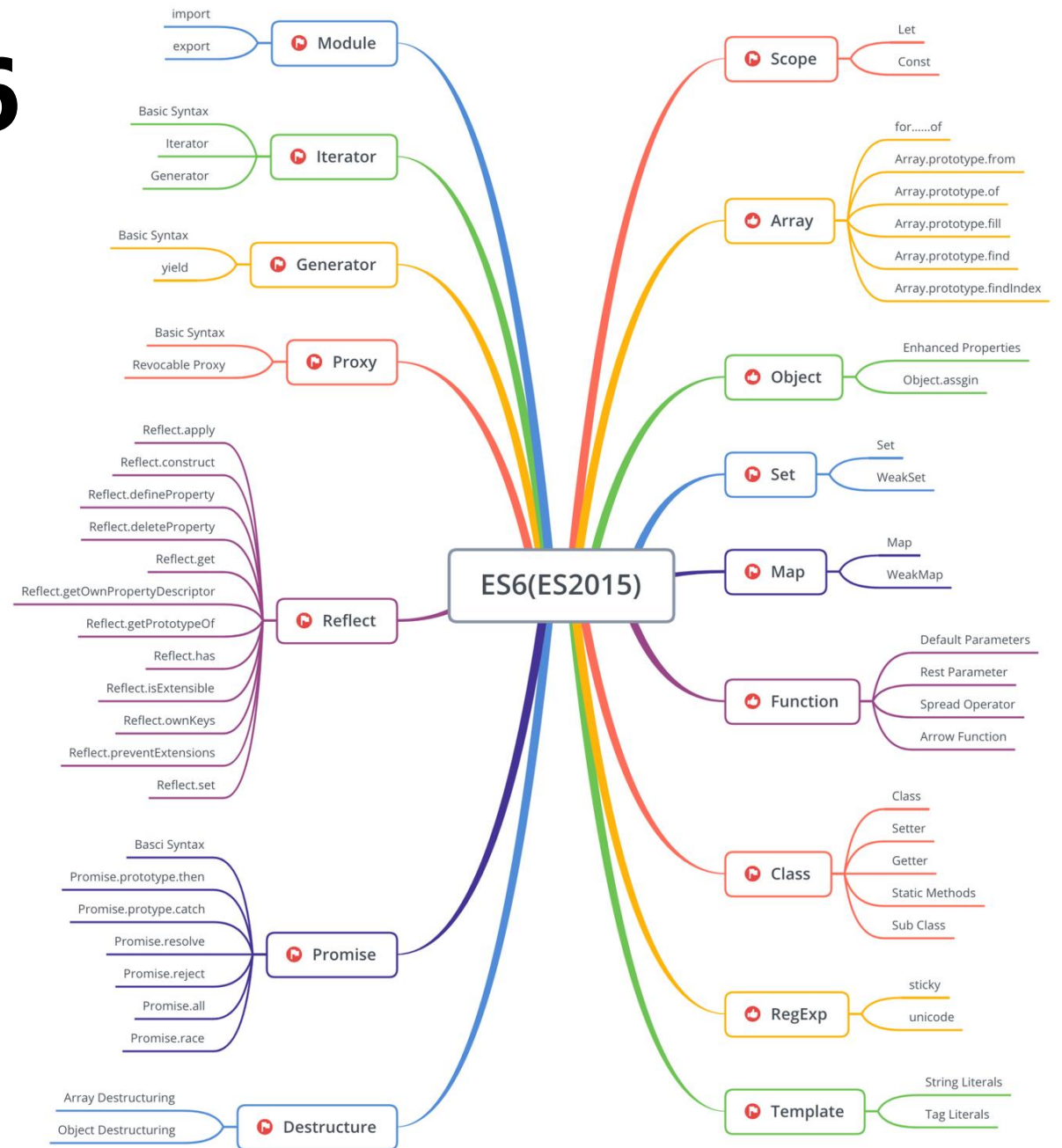
<https://blog.csdn.net/xiaoshixiu>

https://blog.csdn.net/2401_86113453/article/details/142253391

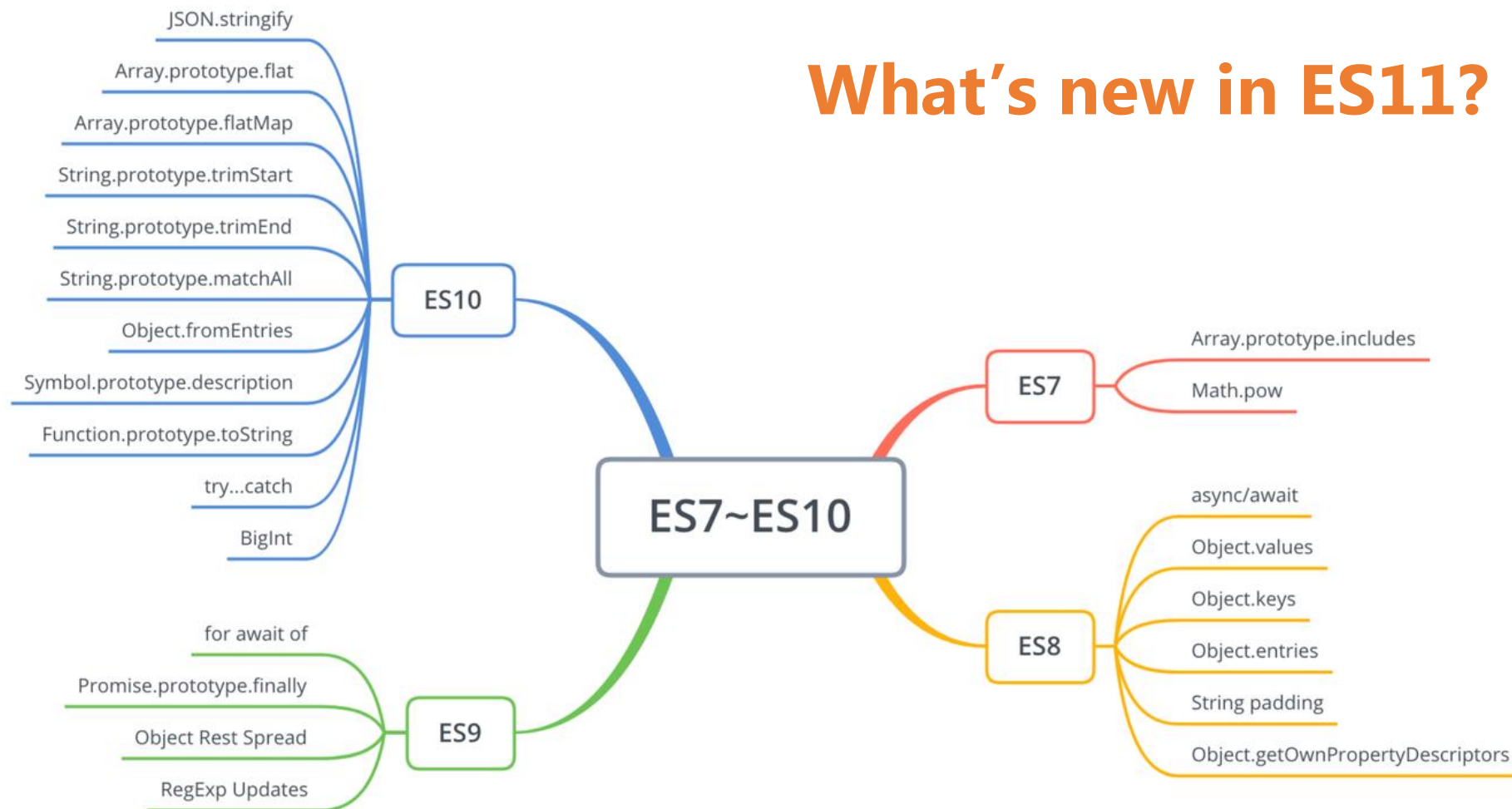
JS Syntax - ES5 vs. ES6



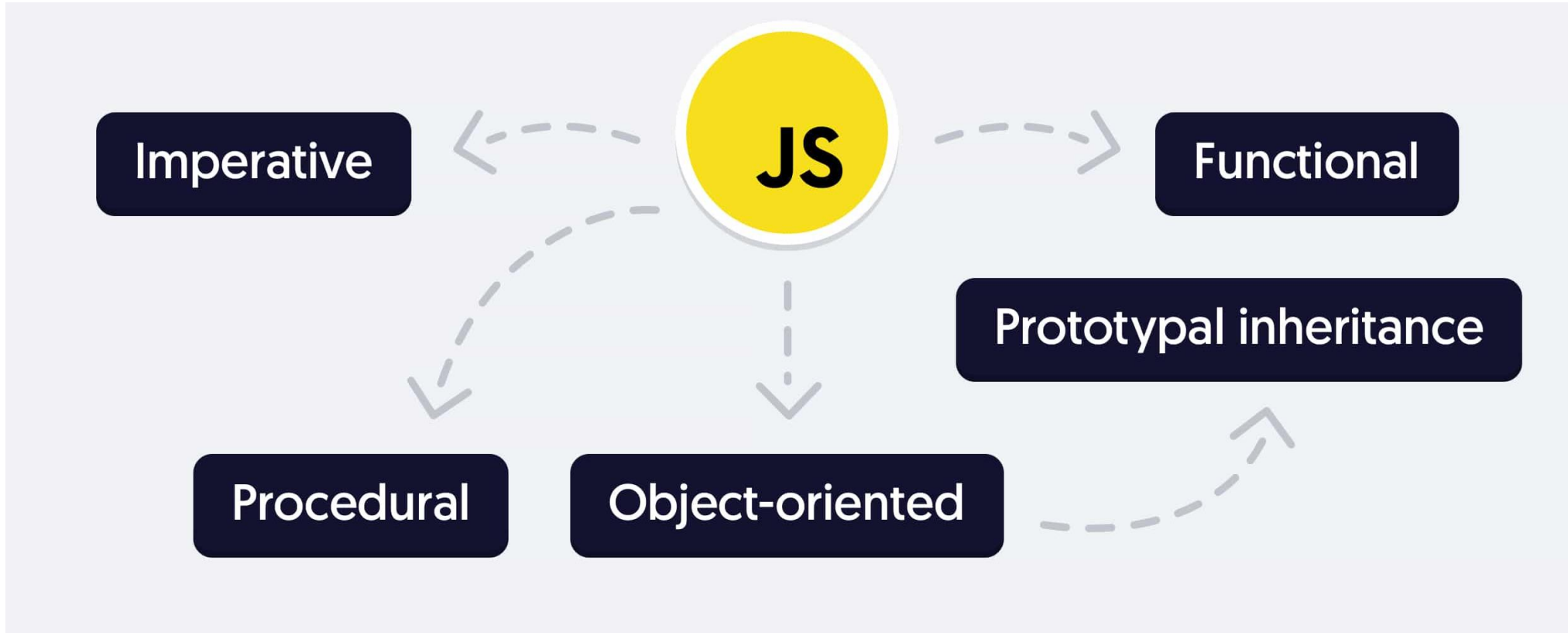
<https://zhuanlan.zhihu.com/p/84738394/>
<https://www.javatpoint.com/es6-generators>



ES7 ~ 10



Paradigm styles

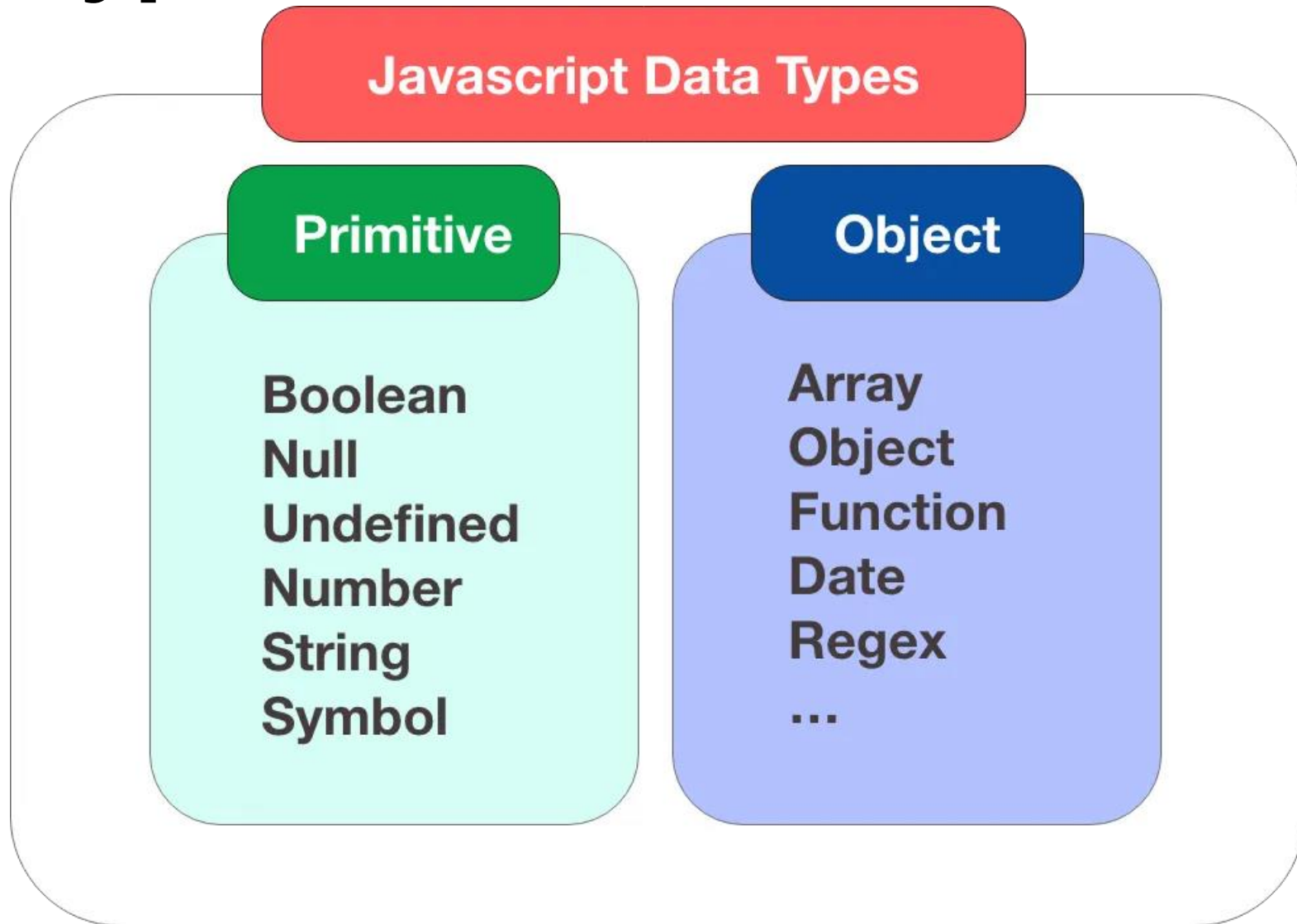


- JavaScript support different paradigms
- Mixing easily causes **confusing**

<https://ultimatecourses.com/blog/an-introduction-to-javascript>

<https://zhuanlan.zhihu.com/p/645486448>

JS data types



Variables in JS



keyword	const	let	var
global scope	NO	NO	YES
function scope	YES	YES	YES
block scope	YES	YES	NO
can be reassigned	NO	YES	YES

- ES5 only has **var**
- ES6 support **let** and **const**

<https://devopsunderground.com/2020/01/es6-variable-declaration-with-let-and-const/>

Variables in JS - const

```
11  
12 var name5 = 'John Adams';  
13 var age5 = 30;  
14 name5 = 'John Hayden';  
15 console.log(name5);
```

John Hayden

```
11  
12 var name5 = 'John Adams';  
13 var age5 = 30;  
14 name5 = 'John Hayden';  
15 console.log(name5);  
16  
17  
18 const name6 = 'John Adams';  
19 var age6 = 30;  
20 name6 = 'John Hayden';  
21 console.log(name6);
```

John Hayden

Uncaught TypeError: Assignment to constant variable.
at script.js:20

- name6 is const, the value cannot be changed.
- This can avoid changing the value by a mistake.

Variables in JS – scope

The screenshot displays a web browser with two functions, `test5` and `test6`, and their execution results in the console.

Function test5 (Left Panel):

```
26 function test5(passed) {
27   if (passed) {
28     var name = 'John';
29     var birthyear = 1990;
30   }
31   console.log(name + ' born in '
32     + birthyear + ' passed the test');
33 }
34 test5(true);
35
```

Function test6 (Right Panel):

```
36 function test6(passed) {
37   let name;
38   const birthyear = 1990;
39   if (passed) {
40     name = 'John';
41   }
42   console.log(name + ' born in '
43     + birthyear + ' passed the test');
44 }
45 test6(true);
46
47
48
49
```

Browser Console (Middle Panel):

The console shows the output of the functions:

- John born in 1990 passed the test
- John born in 1990 passed the test

The console also displays an error message:

```
Uncaught ReferenceError: birthyear is not defined
    at test6 (script.js:42)
    at script.js:44
```

- `let/const` is block scope
- Block means codes in `{ }`

Variables in JS

```
24
25 function test5(passed) {
26   if (passed) {
27     console.log(name);
28     var name = 'John';
29     var birthyear = 1990;
30   }
31   console.log(name + ' born in '
32     + birthyear + ' passed the test');
33 }
34 test5(true);
35
```

Elements Console Sources

top

undefined

John born in 1990 passed the test

```
38
39 function test6(passed) {
40   console.log(name);
41   let name;
42   const birthyear = 1990;
43   if (passed) {
44     name = 'John';
45   }
46   console.log(name + ' born in '
47     + birthyear + ' passed the test');
48 }
49 test6(true);
50
51
52
53
```

top

Uncaught ReferenceError: Cannot access 'name' before initialization
at test6 (script.js:40)
at script.js:49

```
1 function foo(){
2   console.log(name) //undefined
3   var name = 'Jay'
4 }
5 foo()
```



```
1 function foo(){
2   var name
3   console.log(name) //undefined
4   name = 'Jay'
5 }
6 foo()
```

- var Hoisting (变量提升)
- It means can be used before declare.

Variables in JS

```
58 |  
59   var i = 100;  
60   for(var i = 0; i < 5; i++){  
61       console.log(i);  
62   }  
63   console.log(i);  
64  
65  
66  
67
```

```
58 |  
59   let i = 100;  
60   for(let i = 0; i < 5; i++){  
61       console.log(i);  
62   }  
63   console.log(i);  
64  
65  
66  
67  
68
```

- Try them to see the difference

<https://devopsunderground.com/2020/01/es6-variable-declaration-with-let-and-const/>

JS CLASS

- Syntax is similar to JAVA

```
ES6
1 class Employee {
2   constructor(date, month, year) {
3     this.name = 'Employee';
4     this.date = date;
5     this.month = month;
6     this.year = year;}
7   nameRef() {
8     console.log('Name of the class is: ', this.name + '.');
9   }
10 }
11 let e = new Employee(27, 9, 1991);
12 e.nameRef();
13 console.log('Date of birth of the employee ' + e.date + '/' + e.month+ '/' + e.year);
```

Console

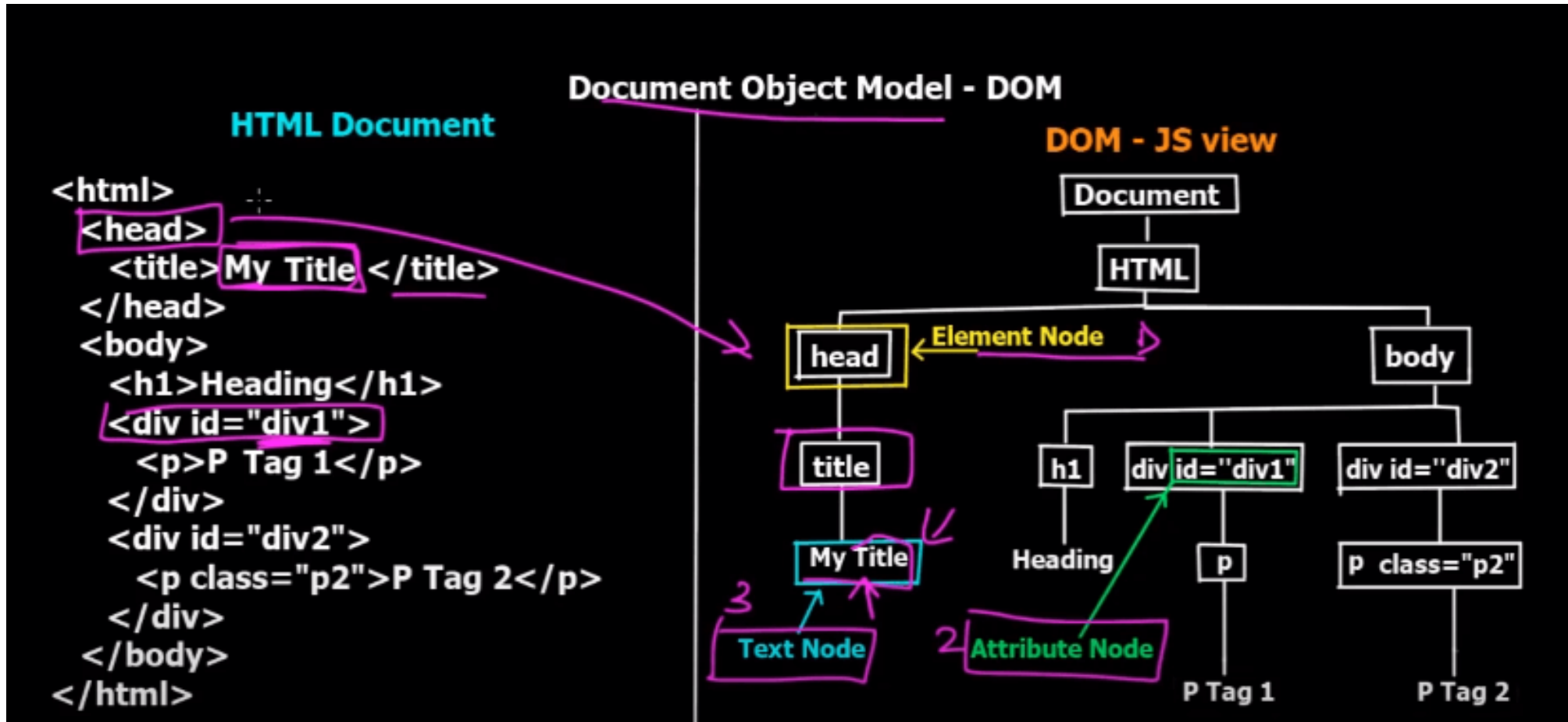
Name of the class is: Employee.

Date of birth of the employee 27/9/1991

<https://www.educba.com/es6-classes/>

<https://blog.polybdenum.com/2017/12/29/how-es6-classes-really-work-and-how-to-build-your-own.html>

JS and DOM



- Different node types

DOM API – select nodes

Gets	Selector Syntax	Method
ID	#demo	getElementById()
Class	.demo	getElementsByClassName()
Tag	demo	getElementsByTagName()
Selector (single)		querySelector()
Selector (all)		querySelectorAll()

```
<h2>ID (#demo)</h2>  
<div id="demo">Access me by ID</div>
```

```
• const demoId = document.getElementById('demo');
```

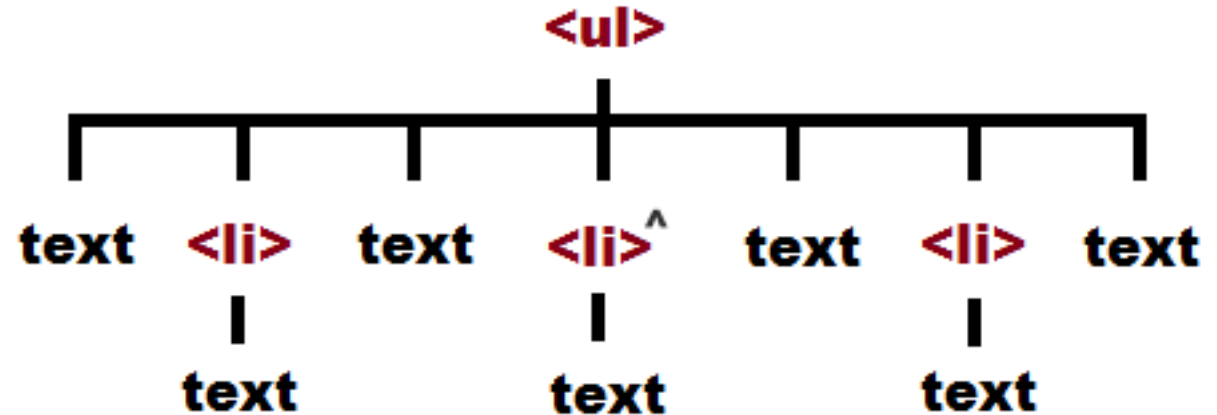
- **A sample to access the div nodeId is “demo”**

<https://mempelajari.com/javascript-get-element-from-node>

<https://learnjavascriptfast.com/chapter-7-the-javascript-document-object-model/3/>

DOM API – select neighbor nodes

```
<ul>
  <li>A</li>
  <li>B</li>
  <li>C</li>
</ul>
```



- `document.querySelector("ul").childNodes[xx]`
- `document.querySelector("ul").firstChild`
- `document.querySelector("ul").lastChild`
- `document.querySelector("ul").childNodes[3].previousSibling`
- `document.querySelector("ul").childNodes[3].nextSibling`
- `document.querySelector("ul").parentNode`

DOM API – get attributes

```
<html>
  <head>
    <title>DOM Practice</title>
  </head>
  <body>
    <div id = "first">The first div tag</div>
    <div class = "myclass">The second div tag</div>
    <div class = "myclass">The third div tag</div>
    <script type="text/javascript" src="chap7.js"></script>
  </body>
</html>
```

The first div tag

The second div tag

The third div tag

```
document.getElementById("first").style.color = "Red";
document.getElementById("first").style.backgroundColor = "Yellow";
```

- style is also an attribute
- You can change font, className ...

DOM API – change contents

```
<html>
  <head>
    <title>DOM Practice</title>
  </head>
  <body>
    <div id = "first">The first div tag</div>
    <div class = "myclass">The second div tag</div>
    <div class = "myclass">The third div tag</div>
    <script type="text/javascript" src="chap7.js"></script>
  </body>
</html>
```

The first div tag
The second div tag
The third div tag

```
var firstDiv = document.getElementById("first");
```

```
firstDiv.textContent = "<em>I've changed</em>";
```

```
firstDiv.innerHTML = "<em>I've changed</em>";
```

- **textContent** will treat `<em>` as the content to be shown
- **innerHTML** will treat `<em>` as a HTML tag

DOM API – change attributes

```
<html>
  <head>
    <title>DOM Practice</title>
  </head>
  <body>
    <div id = "first">The first div tag</div>
    <div class = "myclass">The second div tag</div>
    <div class = "myclass">The third div tag</div>
    <script type="text/javascript" src="chap7.js"></script>
  </body>
</html>
```

The first div tag

The second div tag

The third div tag

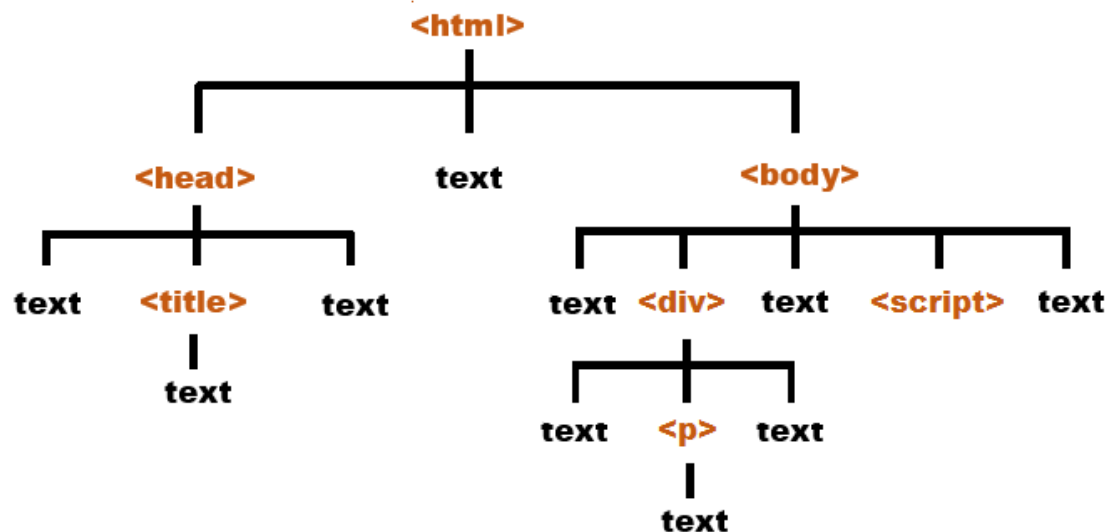
```
document.getElementById("first").style.color = "Red";
document.getElementById("first").style.backgroundColor = "Yellow";
```

- style is also an attribute
- You can change font, className ...

DOM API – insert a node

```
<html>
  <head>
    <title>Nodes Practice</title>
  </head>
  <body>
    <div id = "original">
      <p>Just a simple paragraph</p>
    </div>
    <script type="text/javascript" src="chap9.js"></script>
  </body>
</html>
```

```
var para = document.createElement("p");
var text = document.createTextNode("New Paragraph Added");
para.appendChild(text);
document.querySelector("#original").appendChild(para);
```



Just a simple paragraph

New Paragraph Added

- A new **<p>** was added as the child of **<div>**

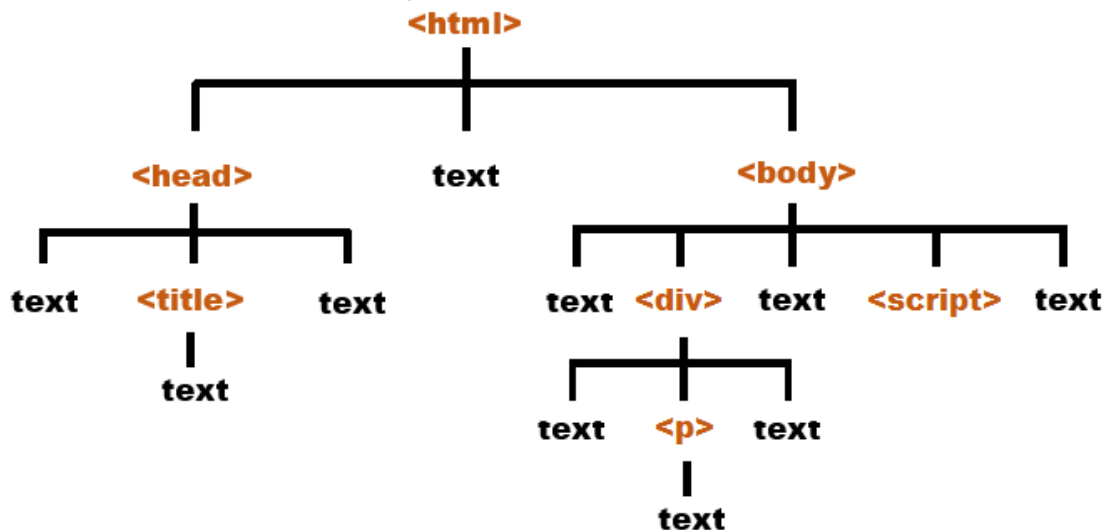
DOM API – delete a node

```
<html>
  <head>
    <title>Nodes Practice</title>
  </head>
  <body>
    <div id = "original">
      <p>Just a simple paragraph</p>
    </div>
    <script type="text/javascript" src="chap9.js"></script>
  </body>
</html>
```

```
var nodeToRemove = document.getElementsByTagName("p")[0];
```

```
removeChild(nodeToRemove); Won't work
```

```
nodeToRemove.parentNode.removeChild(nodeToRemove);
```



- You must operate everything based on an element
- Only can remove a node from its parent

DOM API – event

GeeksforGeeks

HTML DOM ondrop Event

Dragable element

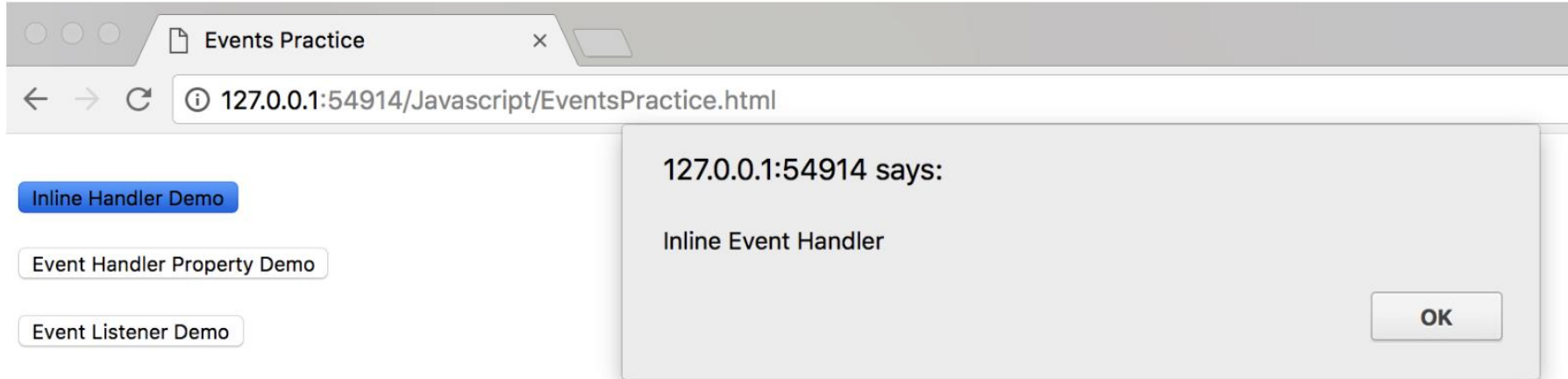


- **Event means something happen**
- **Monitor it and make an action**

<https://vimsky.com/examples/usage/html-dom-ondrop-event.html>

<https://developer.mozilla.org/en-US/docs/Web/API/Event>

DOM API – event → inline handler



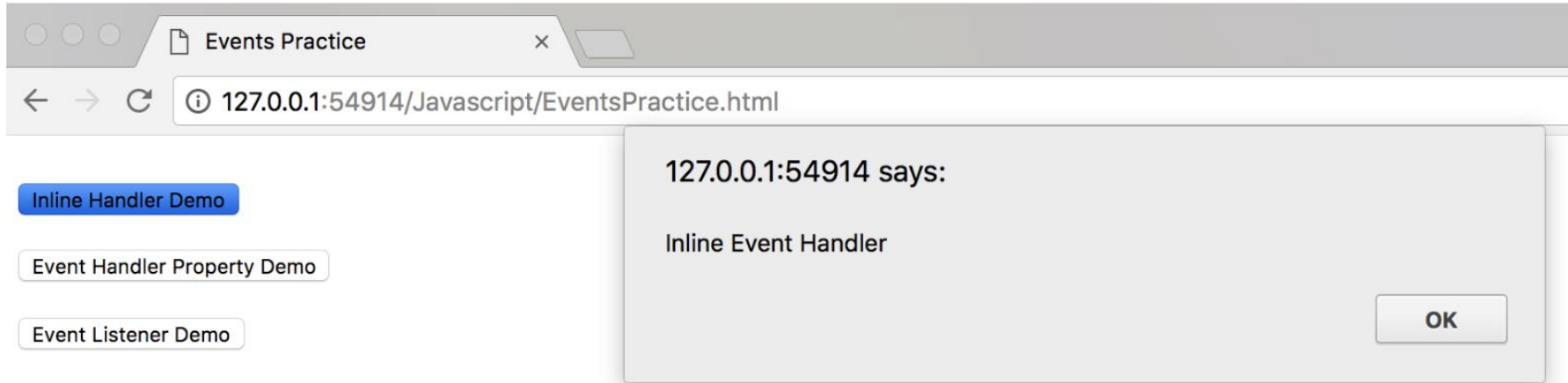
```
<input type = "button" name = "button1" value = "Inline Handler Demo" onclick = "eventHandlerInlineDemo();" >
```

```
function eventHandlerInlineDemo(){  
  alert("Inline Event Handler");  
}
```

- You can write the "onclick" as one attribute with the element

<https://learnjavascriptfast.com/chapter-10-event-handling/3/>

DOM API – event → handler attribute



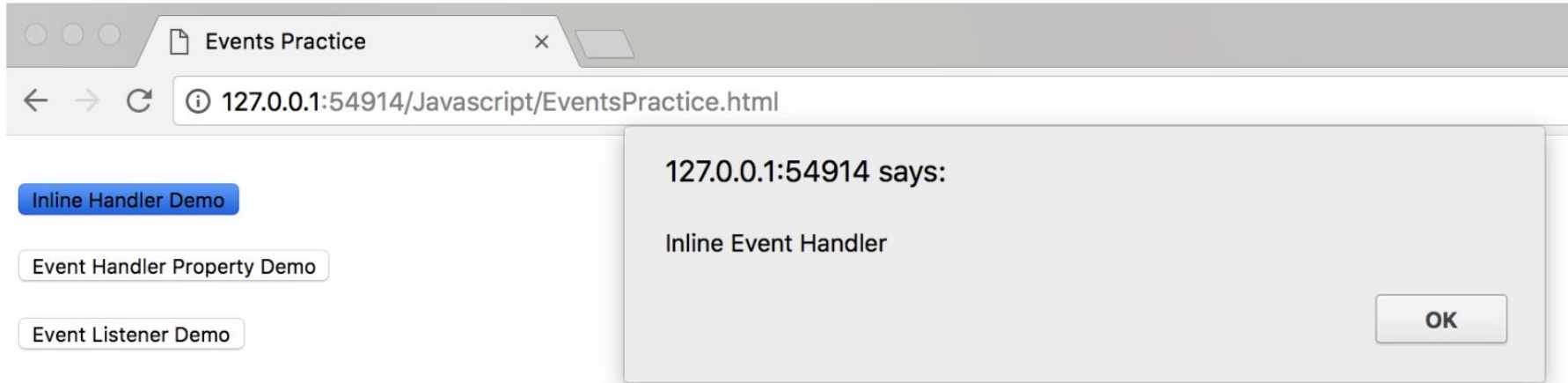
```
<input type = "button" id = "propdemo" name = "button2" value = "Event Handler Property Demo"><br><br>
```

```
document.getElementById("propdemo").onclick = eventHandlerPropDemo;  
document.getElementById("propdemo").onclick = eventHandlerPropDemo2;
```

```
function eventHandlerPropDemo(){  
  alert("Event Handler Property");  
}  
function eventHandlerPropDemo2(){  
  alert("Event Handler Property 2");  
}
```

- You can set/change the “onclick” attribute through the target element
- But one element can only bind one function

DOM API – event → listener



```
<input type = "button" id = "listenerdemo" name = "button3" value = "Event Listener Demo">
```

```
document.getElementById("listenerdemo").addEventListener('click', eventListenerDemo);  
document.getElementById("listenerdemo").addEventListener('click', eventListenerDemo2);
```

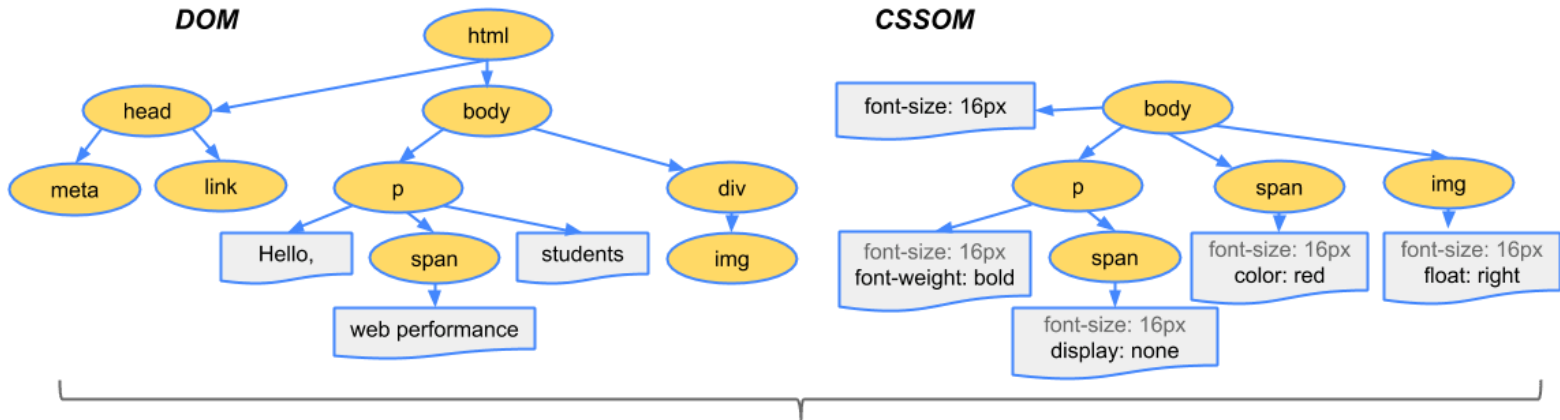
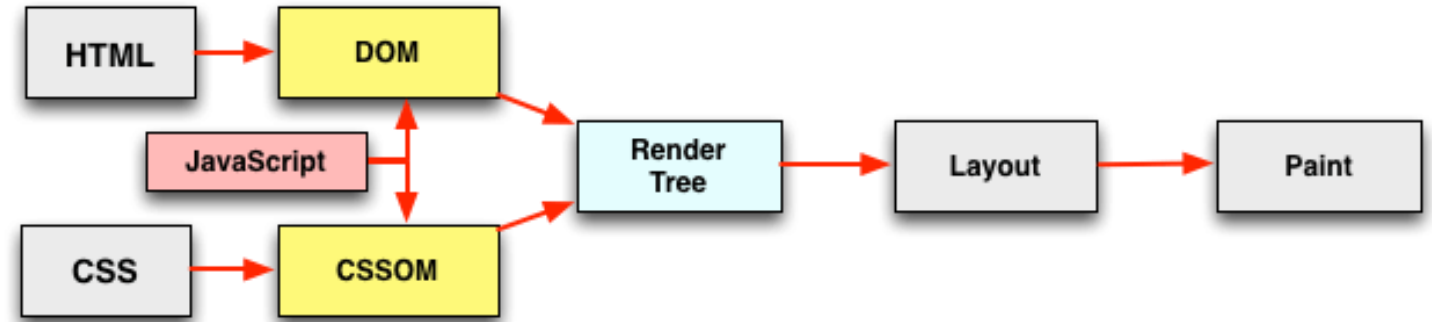
```
function eventListenerDemo(){  
    alert("Event Listener");  
}  
function eventListenerDemo2(){  
    alert("Event Listener 2");  
}
```

- Multiple event handlers can be added to a single HTML element.
- Event listeners can be easily removed using the **removeEventListener()** method

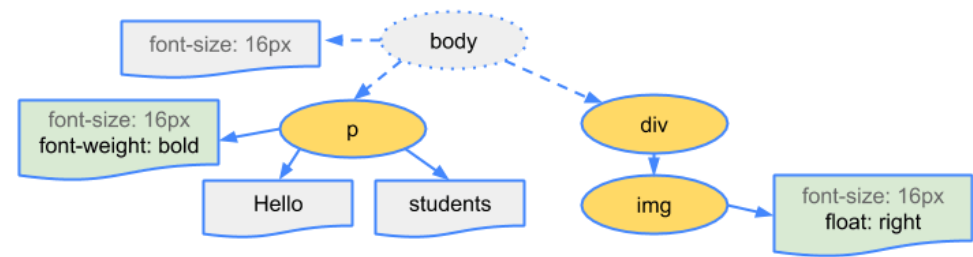
<https://learnjavascriptfast.com/chapter-10-event-handling/3/>

<https://blog.javascripttoday.com/blog/using-event-listeners-in-javascript/>

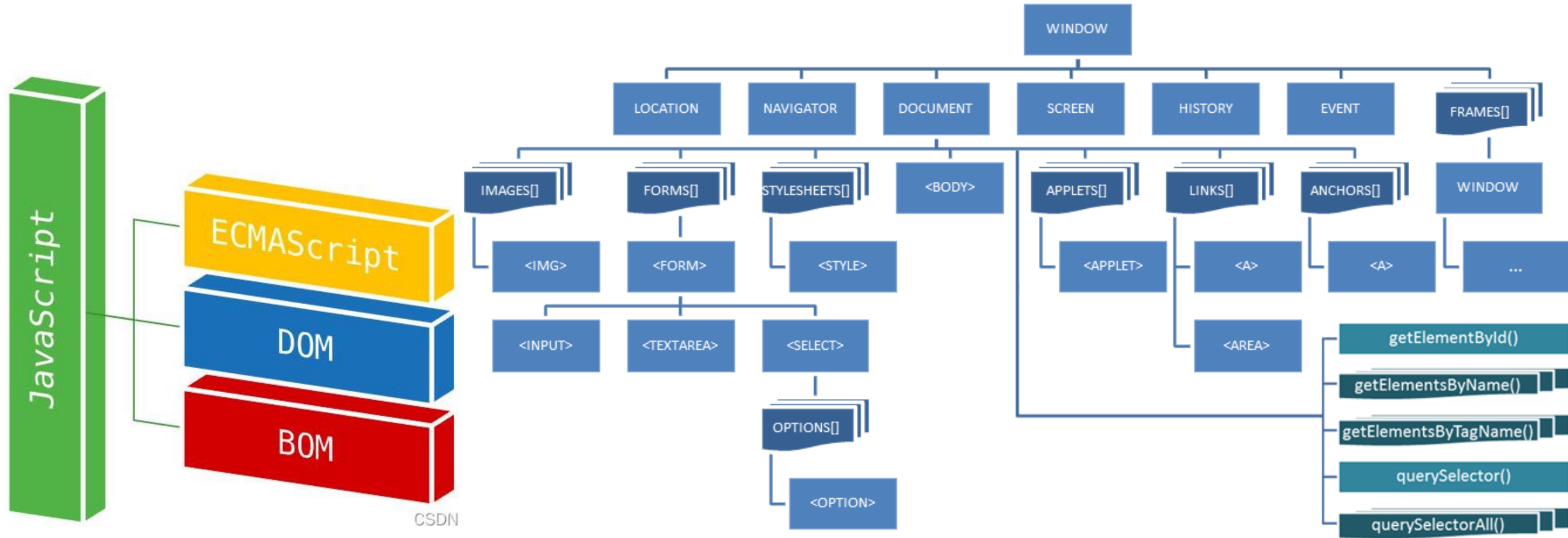
DOM render process



Render Tree



BOM & DOM



- **Browser object model (BOM)** is a JS interface that allows developers to drive the browser
- **Document object model (DOM)** is an interface that allows developers to manipulate the content, structure and style of a website

<https://www.holisticseo.digital/technical-seo/dom/>

BOM API

Window Object

Open a new window when clicking on a button
Open a new window and control its appearance
Blur and Focus a new window
Close the new window
Checks whether the new window has been closed or not
Write some text to the source (parent) window
Move the new window relative to its current position
Move the new window to the specified position
Print the current page
Resize a window by the specified pixels
Resize a window to a specified size
Scroll the content by the specified number of pixels
Scroll the content to a specified position

Location Object

Return the hostname and port of the current URL
Return the entire URL of the current page
Return the path name of the current URL
Return the protocol portion of the current URL
Load a new document
Replace the current document
Break out of a frame

Screen Object

The visitor's screen: Width
The visitor's screen: Height
The visitor's screen: Available Width
The visitor's screen: Available Height
The visitor's screen: Color Depth
The visitor's screen: Pixel Depth

BOM API

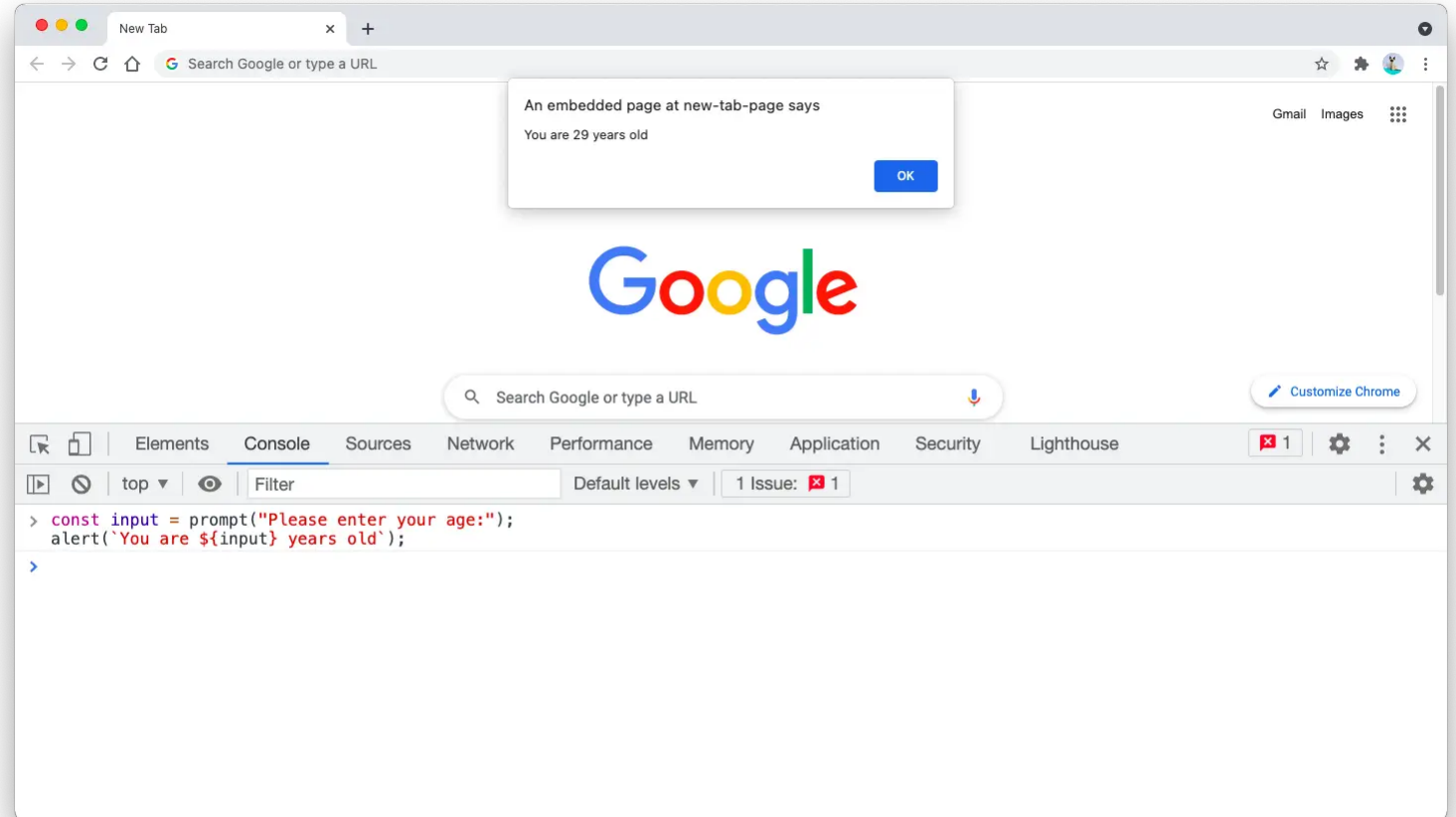
Popup Boxes

Display an alert box

Demonstrate line breaks in an alert box

Display a confirm box

Display a prompt box



- **You should be familiar with this one**

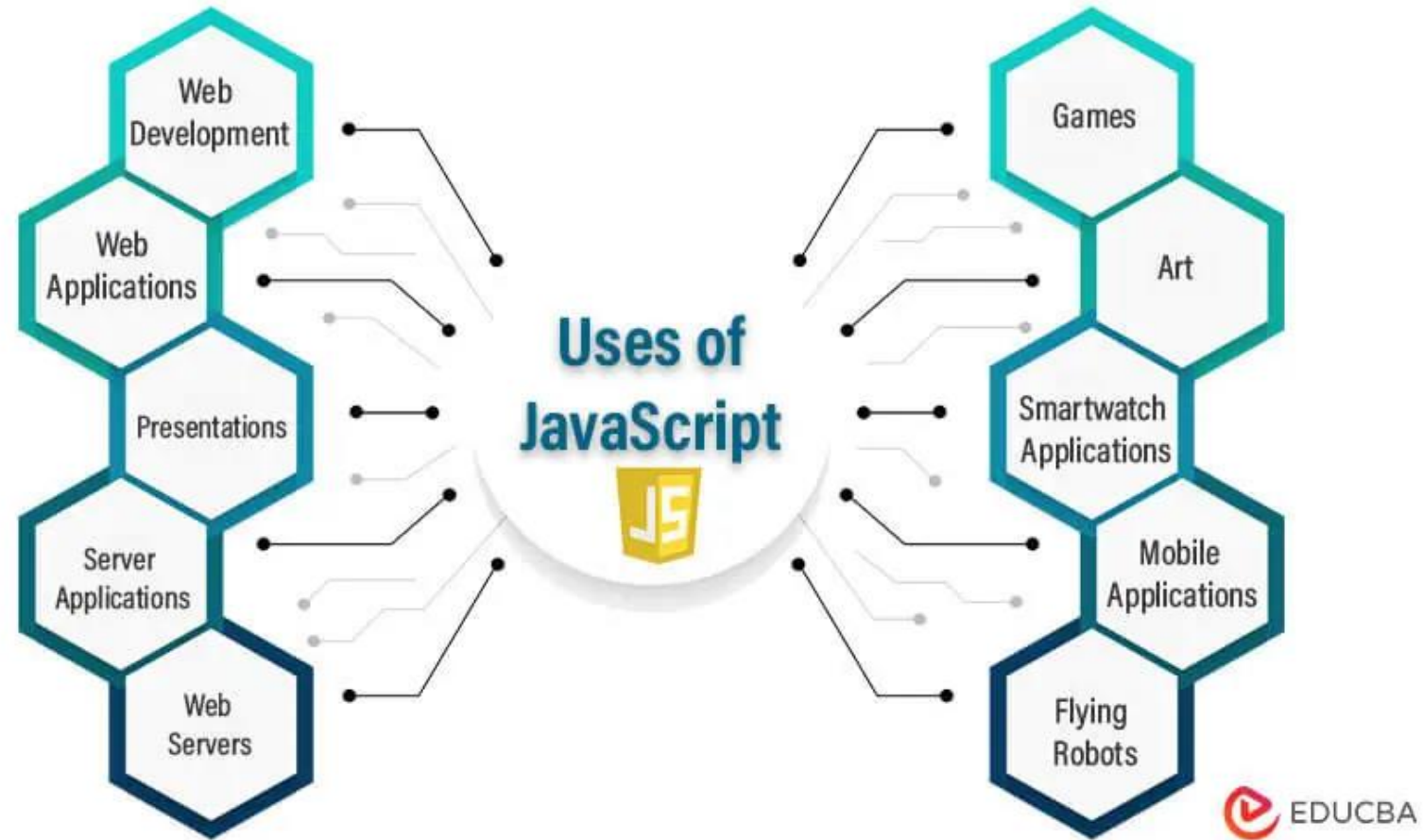
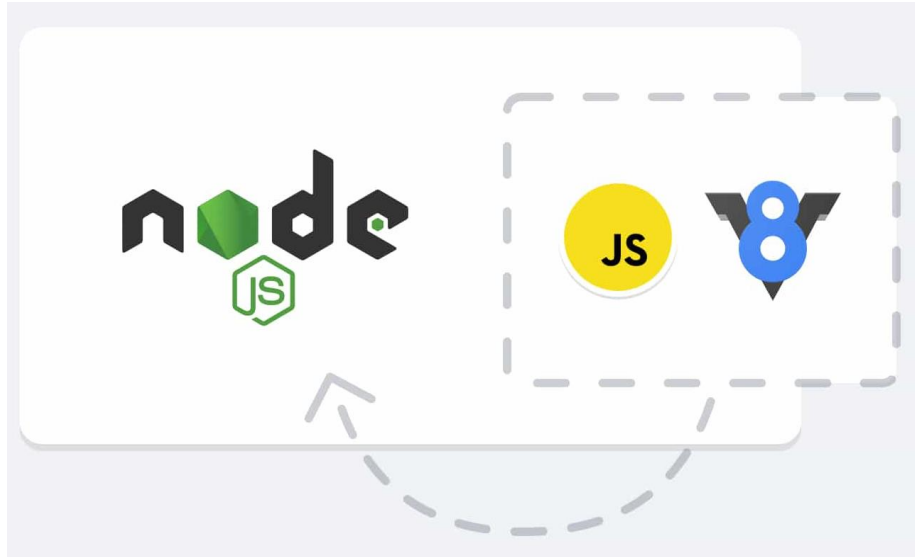
https://www.w3schools.com/js/js_ex_browser.asp

DOM/BOM again

[illegible]

- **You can only learn them through code in demo/true project**

More than WEB



- Since V8 has an excellent performance, it has been implemented in many fields
- NodeJS is an web server that using JS as program language
- For sure, NodeJS does not support BOM/DOM

<https://www.educba.com/uses-of-javascript/>

Don't think it but code it!

```
var arr = [5, 22, 14, 9];
```

```
console.log(arr.sort());
```

// Output

```
[14, 22, 5, 9]
```

> typeof NaN	> true==1
< "number"	< true
> 9999999999999999	> true===1
< 10000000000000000	< false
> 0.5+0.1==0.6	> (!+[[]+[]+![]).length
< true	< 9
> 0.1+0.2==0.3	> 9+"1"
< false	< "91"
> Math.max()	> 91-"1"
< -Infinity	< 90
> Math.min()	> []==0
< Infinity	< true
> []+[]	
< ""	
> []+{}	
< "[object Object]"	
> {}+[]	
< 0	
> true+true+true===3	
< true	
> true-true	
< 0	

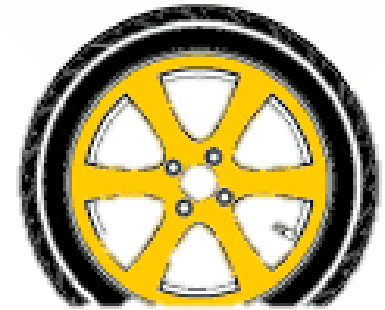


It seems you have everything for front-end???

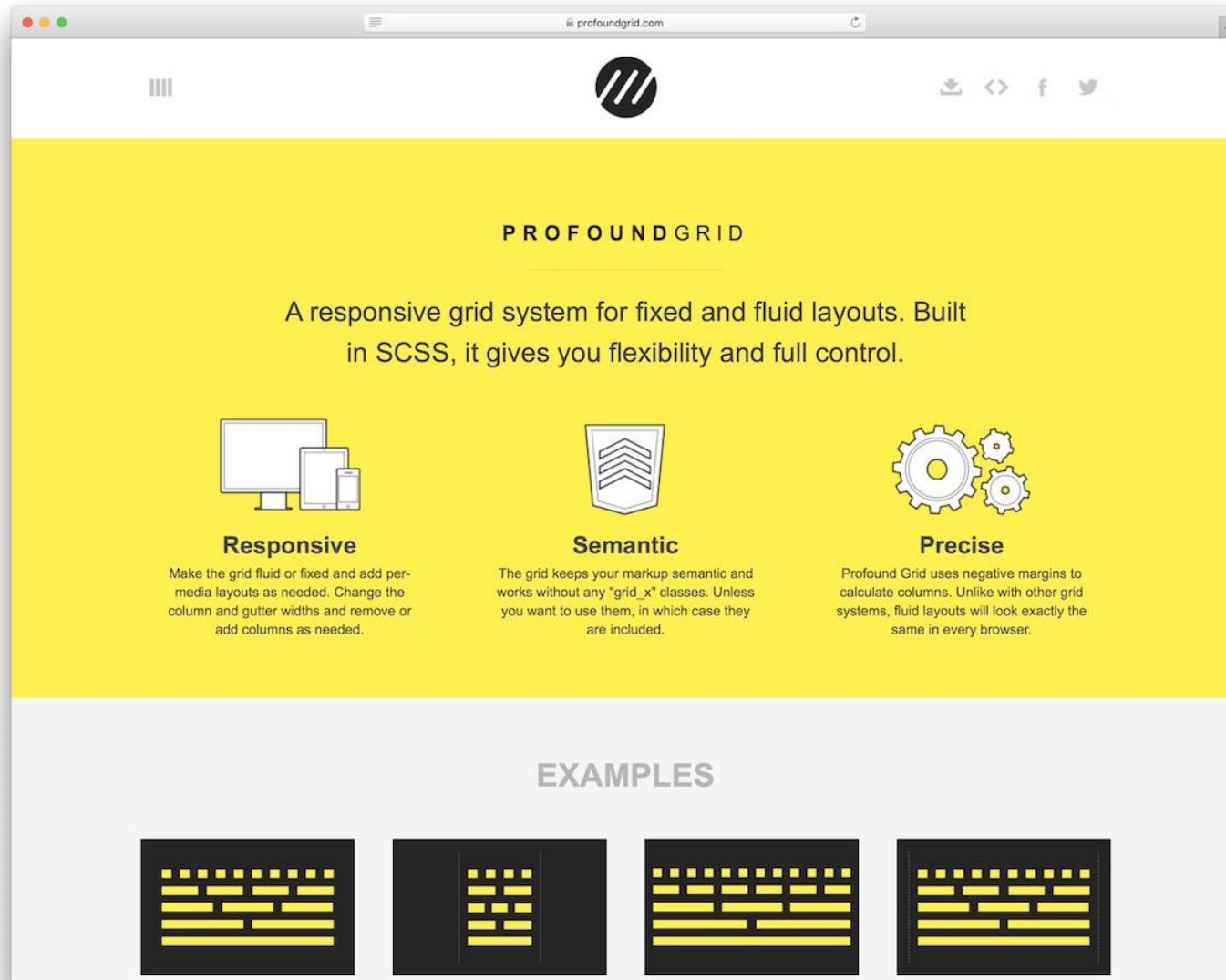
CSS3 JS
HTML5

DON'T REINVENT
THE
WHEEL

Just change the color, size, gears and etc....
NewSteem, Tribes, 8ml, etc....

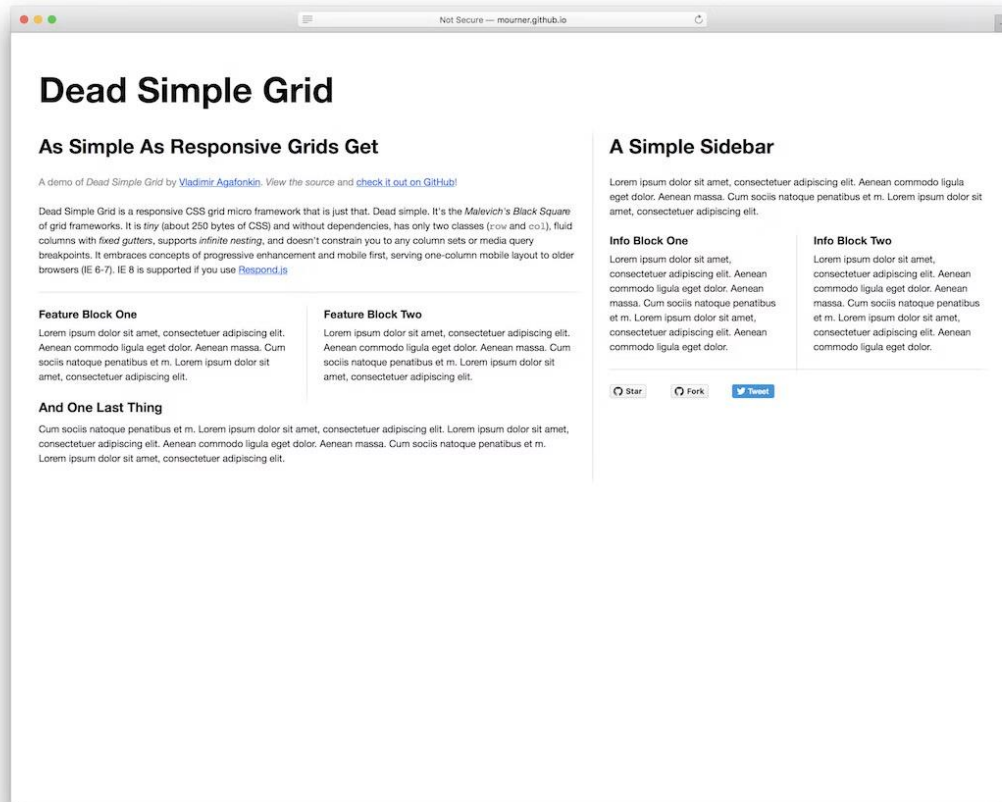


Frameworks do help a lot!



<https://colorlib.com/wp/html5-css3-grid-frameworks/>

Frameworks do help a lot!



<https://colorlib.com/wp/html5-css3-grid-frameworks/>

Bootstrap



- Original from **Twitter** (now **X**).
- Quickly design and customize **responsive** mobile-first sites with Bootstrap, the world's most popular front-end **open source toolkit**, featuring Sass variables and mixins, responsive grid system, extensive prebuilt components, and powerful JavaScript plugins.
- Many **pre-defined classes** in Bootstrap includes CSS, **fonts** and JS (earlier version use jQuery).
- With Bootstrap, develop the front-end project can be much easier.

<https://getbootstrap.com/>

Bootstrap – basic templet to include it

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <!-- The above 3 meta tags *must* come first in the head; any other head content must come
*after* these tags -->
    <title>Bootstrap 101 Template</title>

    <!-- Bootstrap -->
    <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@3.4.1/dist/css/bootstrap.min.css" integrity="sha384-
HSMxcRTRxnN+Bdg0JdbxYKrThec0KuH5zCYotlSAcp1+c8xmyTe9GYg1l9a69psu" crossorigin="anonymous">

    <!-- HTML5 shim and Respond.js for IE8 support of HTML5 elements and media queries -->
    <!-- WARNING: Respond.js doesn't work if you view the page via file:// -->
    <!--[if lt IE 9]>
      <script src="https://cdn.jsdelivr.net/npm/html5shiv@3.7.3/dist/html5shiv.min.js"></script>
      <script src="https://cdn.jsdelivr.net/npm/respond.js@1.4.2/dist/respond.min.js"></script>
    <![endif]-->
  </head>
  <body>
    <h1>Hello, world!</h1>

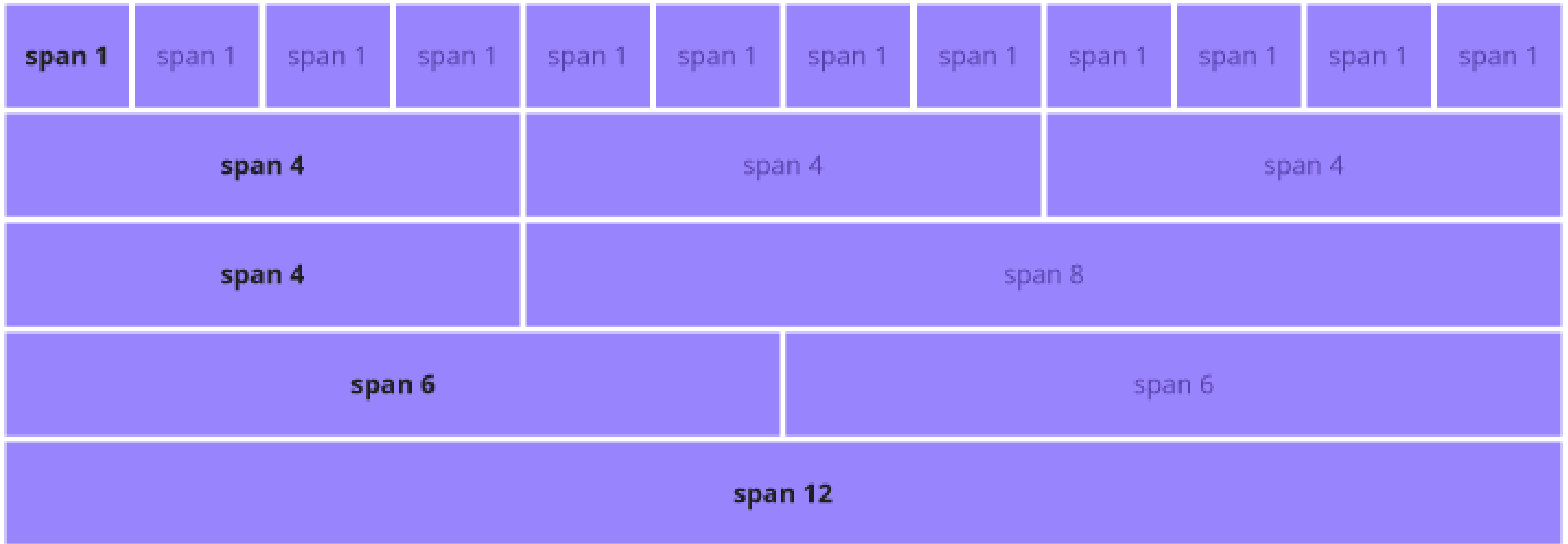
    <!-- jQuery (necessary for Bootstrap's JavaScript plugins) -->
    <script src="https://code.jquery.com/jquery-1.12.4.min.js" integrity="sha384-
nvAa0+6Qg9cLwYCGGPpDQLVpLNN0fRaR0jHqs13t4Ggj3Ez50XnGQqc/r8MhnRDZ" crossorigin="anonymous">
</script>
    <!-- Include all compiled plugins (below), or include individual files as needed -->
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@3.4.1/dist/js/bootstrap.min.js"
integrity="sha384-aJ210j lMXNL5UyIl/XNwTMqvzeRMZH2w8c5crVPzpU8Y5bApTppSuUkhZXN0VxHd"
crossorigin="anonymous"></script>
  </body>
</html>
```

- Bootstrap is actually 1 CSS file + 1 JS file
- “min” means remove all space to make the file as small as possible. It is not friendly for reading.
- You can download them to your web server and use <link> and <script> to include them to the HTML
- You can also use the on-line version (Browser will download them from the specified server).

https://documents.sessions.edu/eforms/courseware/coursedocuments/dreamweaver_ii/bootstrap-cdn.html

<https://getbootstrap.com/docs/3.4/getting-started/>

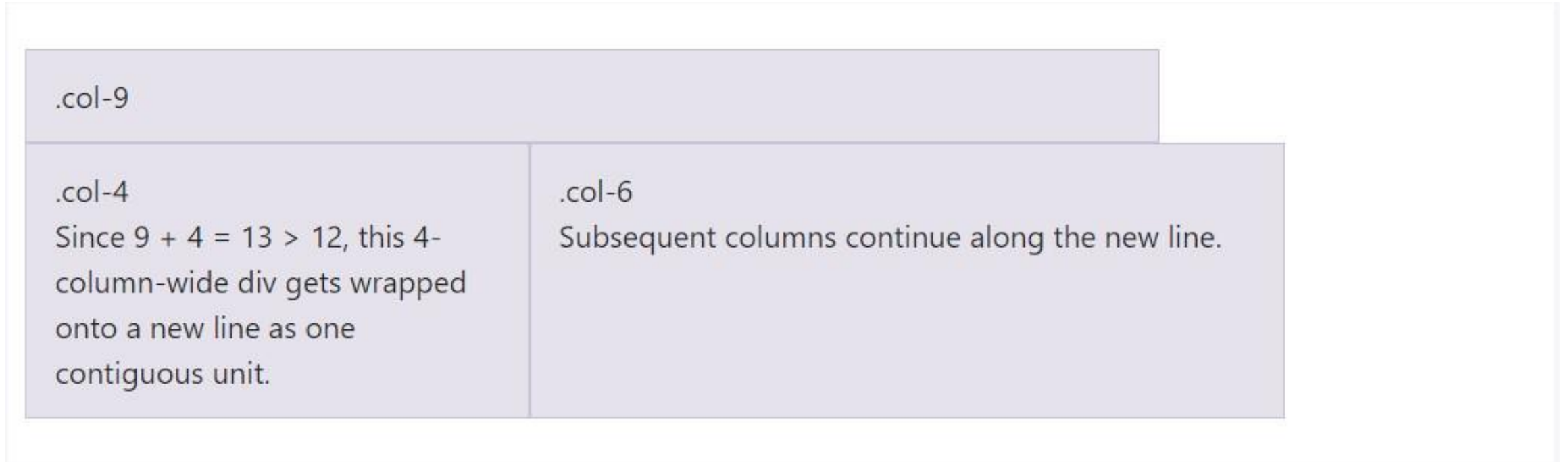
Bootstrap – grid system



- Bootstrap uses a grid system for layout
- Each row has 12 columns
- Each div can take 1-12 columns to achieve the desired layout

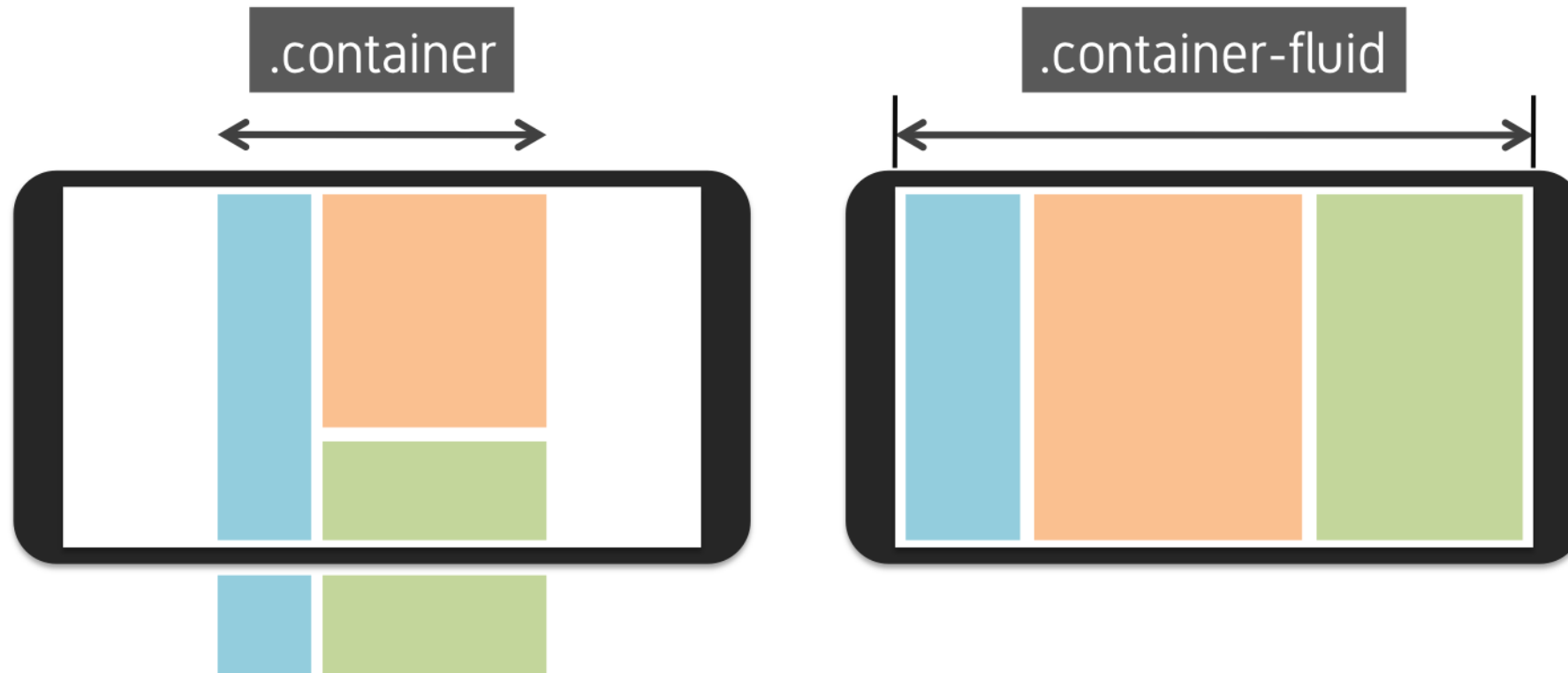
<https://www.bitdegree.org/learn/bootstrap-grid-system>

Bootstrap – grid system



- Since $9 + 4 = 13 > 12$, this 4-column-wide div gets wrapped onto a new line as one contiguous unit.
- Subsequent columns continue along the new line.

Bootstrap – grid system

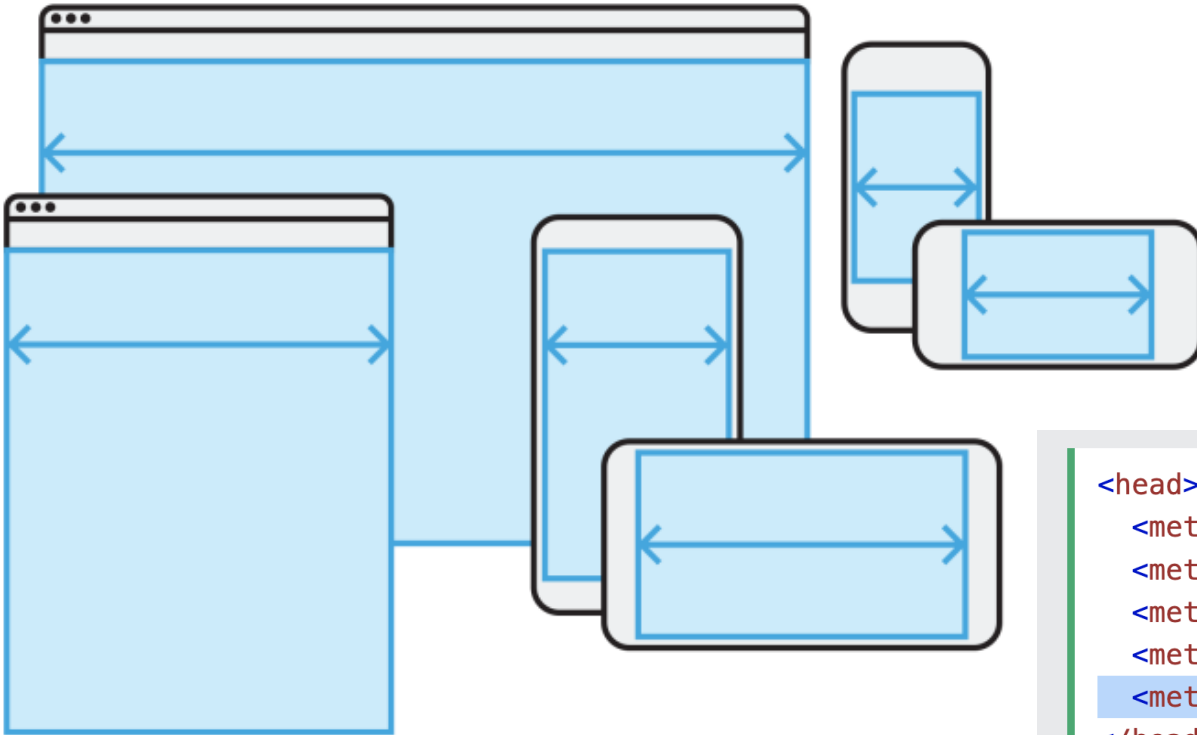


- All elements should be within a container.
- Class `.container` means the absolute width (px).
- Class `.container-fluid` means the relative width (100% by default)

<https://www.bitdegree.org/learn/bootstrap-grid-system>

Bootstrap – breakpoint

100vw = 100% of viewport width



```
<head>
  <meta charset="UTF-8">
  <meta name="description" content="Free Web tutorials">
  <meta name="keywords" content="HTML, CSS, JavaScript">
  <meta name="author" content="John Doe">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
```

- To active the responsive function, <meta> viewport should be specified within the <head>

https://www.w3schools.com/css/css_rwd_viewport.asp

Bootstrap – breakpoint

	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	Extra large ≥1200px
Grid behavior	Horizontal at all times	Collapsed to start, horizontal above breakpoints			
Max container width	None (auto)	540px	720px	960px	1140px
Class prefix	.col-	.col-sm-	.col-md-	.col-lg-	.col-xl-
# of columns	12				
Gutter width	30px (15px on each side of a column)				
Nestable	Yes				
Offsets	Yes				
Column ordering	Yes				

Bootstrap 3 3.3.7

```
@media(max-width:767px){/* до указанного разрешения */}  
@media(min-width:768px){/* после указанного разрешения */}  
@media(min-width:992px){/* после указанного разрешения */}  
@media(min-width:1200px){/* после указанного разрешения */}
```

Bootstrap 4 4.0.0 Beta 2

```
/* До 575px значения наследуются из bootstrap.css */  
  
@media(min-width:576px){/* после указанного разрешения */}  
@media(min-width:768px){/* после указанного разрешения */}  
@media(min-width:992px){/* после указанного разрешения */}  
@media(min-width:1200px){/* после указанного разрешения */}
```

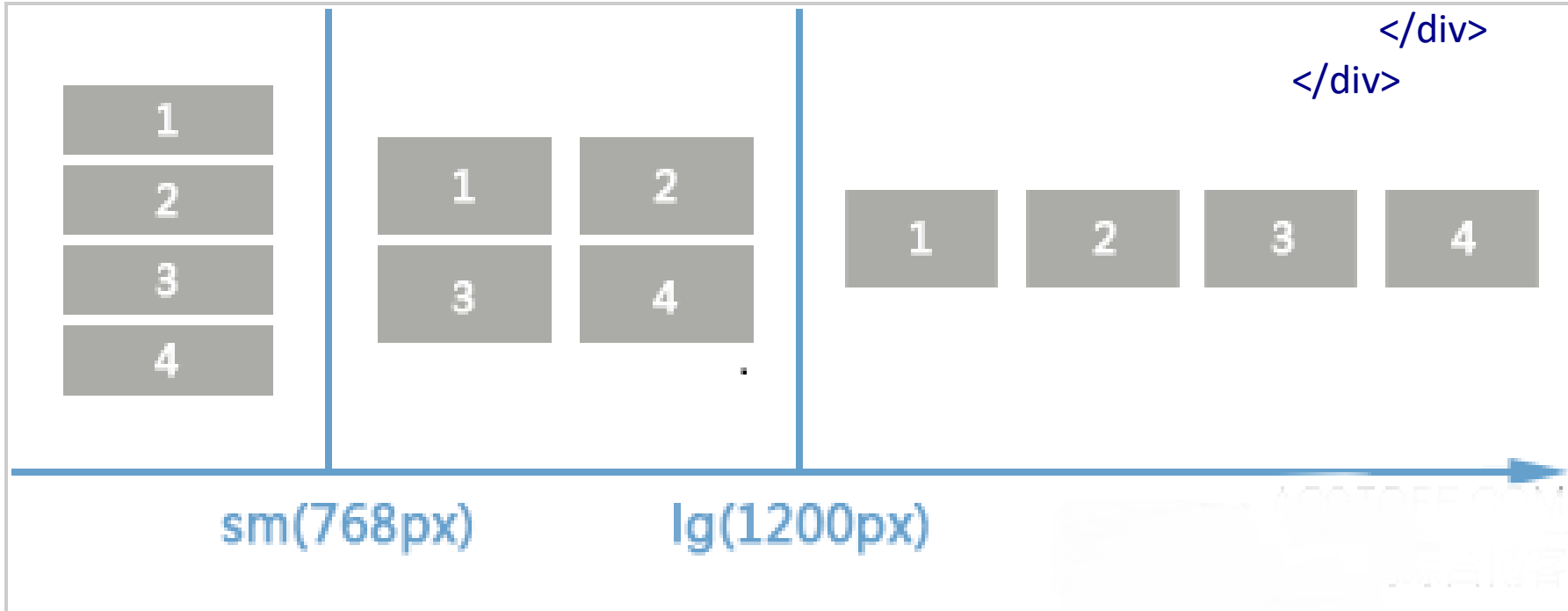
- Breakpoints are customizable widths that determine how your responsive layout behaves across device or viewport sizes in Bootstrap.
- Bootstrap uses media queries (@media) of CSS3

<https://apycom.com/bootstrap-components/bootstrap-grid-318.html>

https://www.drupal.org/project/vartheme_bs4/issues/3051938

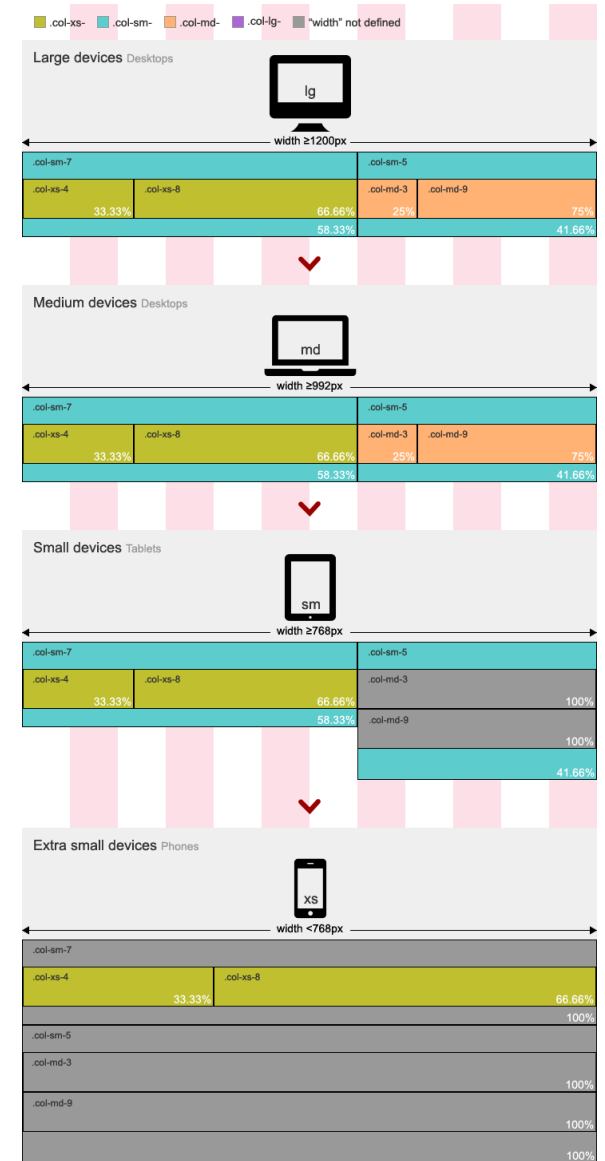
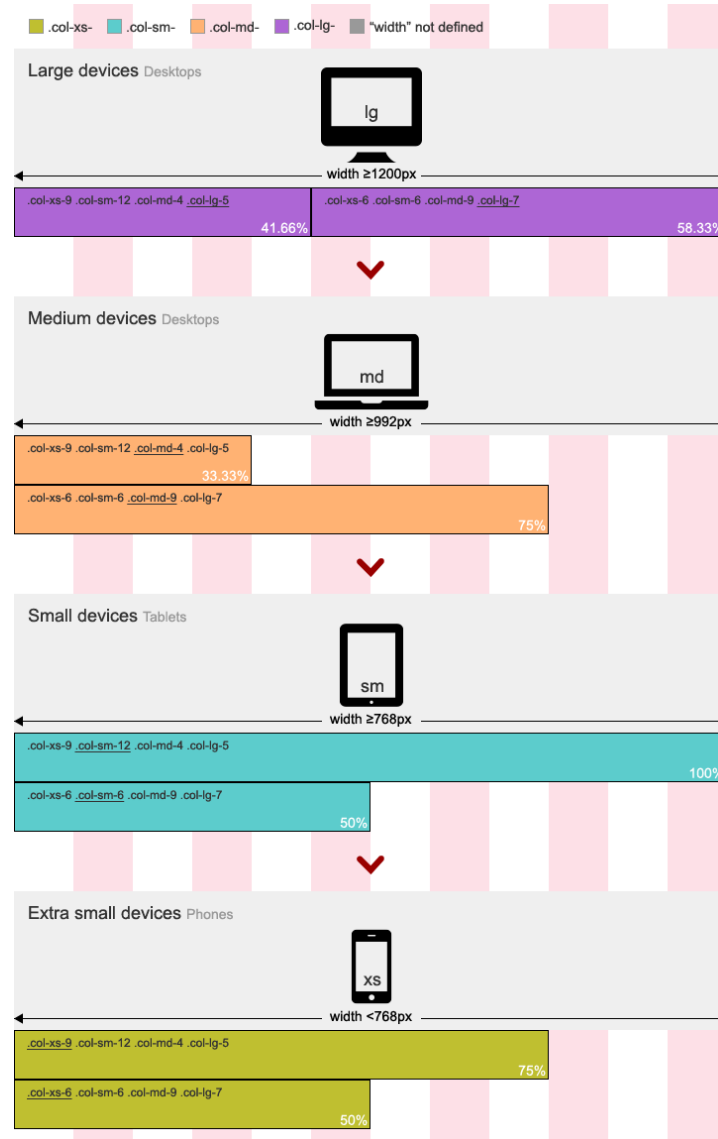
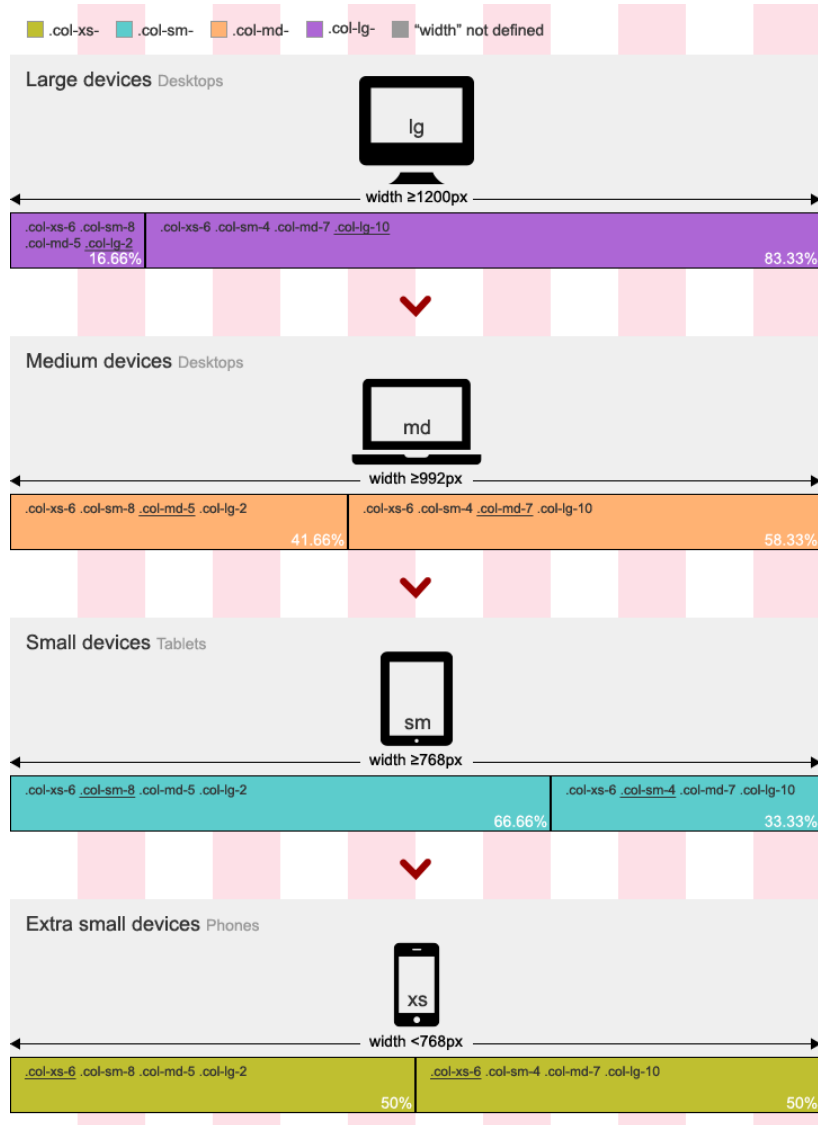
Bootstrap – breakpoint

```
<div class="container">  
  <div class="row">  
    <div class="col-sm-6 col-lg-3">1</div>  
    <div class="col-sm-6 col-lg-3">2</div>  
    <div class="col-sm-6 col-lg-3">3</div>  
    <div class="col-sm-6 col-lg-3">4</div>  
  </div>  
</div>
```



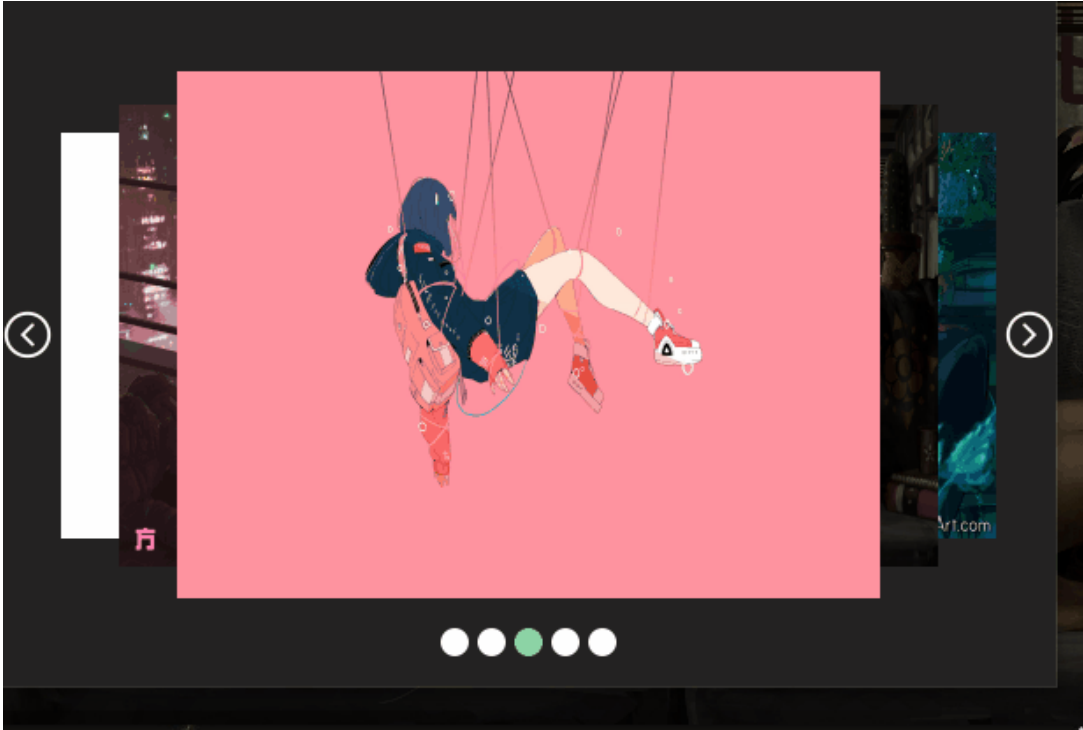
- More than one class can be assigned to one element
- Multiple class priorities won't be discussed here

Bootstrap – breakpoint



<https://www.zeninvader.com/bootstrap-3-grid-system-explained/>

Bootstrap – still a lot ...



Normal Size



Large Size Buttons



Small Size



Extra small Size



Block level buttons



- Carousel

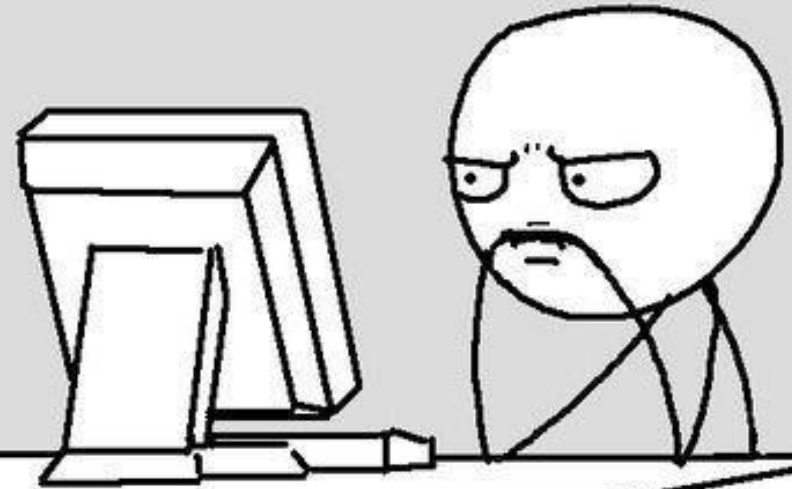
- Buttons

- **Explore more through your labs**

Summary

- An introduction to JS
- An introduction to Bootstrap

It doesn't work..... why?



It works..... why?

